

- 30.** A compound microscope is an optical instrument used for observing highly magnified images of tiny objects. It consists of two convex lenses. The lens near the object is called the objective and the lens near the eye is called the eyepiece. The magnifying power of a compound microscope is given by $m = m_o m_e$, where m_o is the magnification produced by objective lens and m_e is the magnification produced by eyepiece. The expression for m_e depends on whether the final image is formed at the near point or at infinity (normal adjustment).

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- (i) (a) A compound microscope consists of an objective lens of focal length $f_o = 2.0$ cm and an eyepiece of focal length $f_e = 6.25$ cm separated by a distance of 15 cm. How far should an object be placed from the objective so that the final image is formed at infinity ?

- | | |
|-------------|-------------|
| (A) 3.45 cm | (B) 5 cm |
| (C) 1.29 cm | (D) 2.59 cm |

- 30.** A compound microscope is an optical instrument used for observing highly magnified images of tiny objects. It consists of two convex lenses. The lens near the object is called the objective and the lens near the eye is called the eyepiece. The magnifying power of a compound microscope is given by $m = m_o m_e$, where m_o is the magnification produced by objective lens and m_e is the magnification produced by eyepiece. The expression for m_e depends on whether the final image is formed at the near point or at infinity (normal adjustment).

CBSE 2025 C

- (i) (a) A compound microscope consists of an objective lens of focal length $f_o = 2.0$ cm and an eyepiece of focal length $f_e = 6.25$ cm separated by a distance of 15 cm. How far should an object be placed from the objective so that the final image is formed at infinity ?

1

- | | | |
|-------------|-------------|-------------|
| (A) 3.45 cm | (B) 5 cm | |
| (C) 1.29 cm | (D) 2.59 cm | (D) 2.59 cm |

(b) A person with normal near point 25 cm, using a compound microscope with an objective of focal length 0.8 cm and eyepiece of focal length 2.5 cm can bring an object placed at 0.9 cm from the objective in sharp focus. If the image formed by the objective is at a distance of 2.27 cm from the eyepiece, the separation between the lenses is :

1

(A) 7.47 cm

(B) 8.47 cm

(C) 9.47 cm

(D) 10.47 cm

(b) A person with normal near point 25 cm, using a compound microscope with an objective of focal length 0.8 cm and eyepiece of focal length 2.5 cm can bring an object placed at 0.9 cm from the objective in sharp focus. If the image formed by the objective is at a distance of 2.27 cm from the eyepiece, the separation between the lenses is :

1

(A) 7.47 cm

(B) 8.47 cm

(C) 9.47 cm

(D) 10.47 cm

(C) 9.47 cm

- (ii) A compound microscope consists of an objective of focal length 2 cm and an eyepiece of focal length 10 cm. The eyepiece is kept 20 cm away from the objective. Find the magnifying power of the microscope if the final image is formed at infinity.

(A) 7

(B) 10

(C) 14

(D) 16

- (ii) A compound microscope consists of an objective of focal length 2 cm and an eyepiece of focal length 10 cm. The eyepiece is kept 20 cm away from the objective. Find the magnifying power of the microscope if the final image is formed at infinity.

(A) 7

(B) 10

(C) 14

(D) 16

(B) 10

(iii) The intermediate image formed in a compound microscope is : *1*

- (A) real, inverted and magnified
- (B) real, erect and magnified
- (C) virtual, erect and magnified
- (D) virtual, inverted and magnified

(iii) The intermediate image formed in a compound microscope is : *1*

- (A) real, inverted and magnified
- (B) real, erect and magnified
- (C) virtual, erect and magnified
- (D) virtual, inverted and magnified

(A) Real inverted and magnified.

(iv) The focal lengths of the objective and the eyepiece of a compound microscope are f_o and f_e respectively. Which of the following statements is correct, in normal adjustment ?

1

- (A) The object is at a distance less than f_o from the objective.
- (B) The object is at a distance more than f_o and less than $2f_o$ from the objective.
- (C) The intermediate image is formed at a distance slightly more than f_e from the eyepiece.
- (D) The intermediate image is formed at $2f_e$ from the eyepiece.

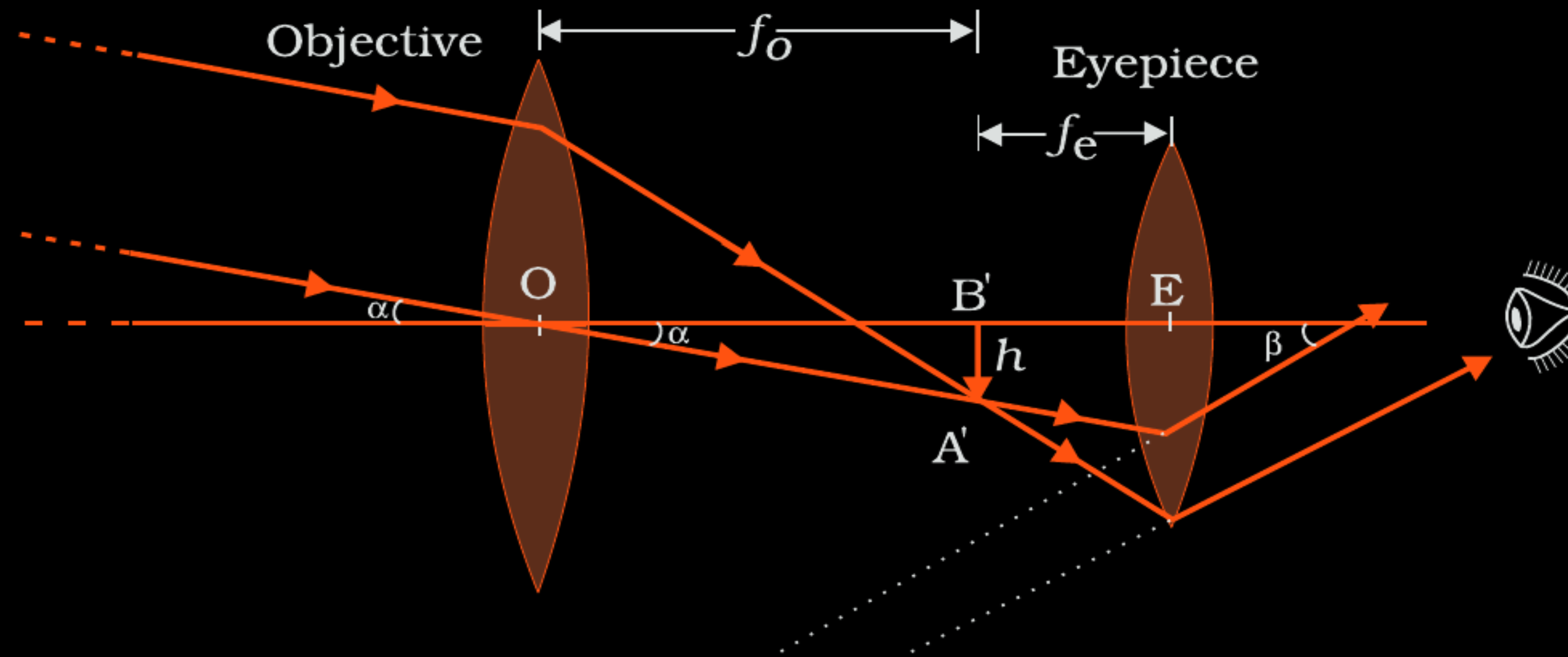
(iv) The focal lengths of the objective and the eyepiece of a compound microscope are f_o and f_e respectively. Which of the following statements is correct, in normal adjustment ? 1

- (A) The object is at a distance less than f_o from the objective.
- (B) The object is at a distance more than f_o and less than $2f_o$ from the objective.
- (C) The intermediate image is formed at a distance slightly more than f_e from the eyepiece.
- (D) The intermediate image is formed at $2f_e$ from the eyepiece.

(B) The object is at a distance more than f_o and less than $2 f_o$ from the objective

- 32.** (a) (i) Explain with the help of a labelled ray diagram the formation of final image by an astronomical telescope at infinity. Write the expression for its magnifying power.
- (ii) The total magnification produced by a compound microscope is 20. The magnification produced by the eyepiece is 5. When the microscope is focussed on a certain object, the distance between the objective and eyepiece is observed to be 14 cm. Calculate the focal lengths of the objective and the eyepiece. (Given that the least distance of distinct vision = 25 cm)

5



(Note: Deduct ½ mark, for not showing arrows with the rays)

Light from distant object enters the objective lens & forms a real image A'B' at f_o .

This image A'B' acts as an object for eye piece and eye piece forms a magnified image at infinity.

$$\text{Magnifying Power} = \frac{f_o}{f_e}$$

1½

½

1

(ii) Image is formed at least distance of distinct vision $20 = m_o \times m_e$ $m_o = \frac{20}{5} = 4$ $m_e = 1 + \frac{D}{f_e}$ $f_e = \frac{25}{4} \text{ cm}$ $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$ $\frac{1}{-25} - \frac{1}{u_e} = \frac{4}{25}$ $u_e = -5 \text{ cm}$ $L = v_o + u_e $ $v_o = 9 \text{ cm}$ Given, $\frac{v_o}{u_o} = 4$ $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$ $\frac{1}{f_o} = \frac{1}{9} - \left(-\frac{4}{9}\right)$ $f_o = \frac{9}{5} \text{ cm}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
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15. *Assertion (A) :* In a reflecting telescope, the image does not have chromatic aberration.

Reason (R) : Chromatic aberration occurs only due to refraction of light through an optical medium.

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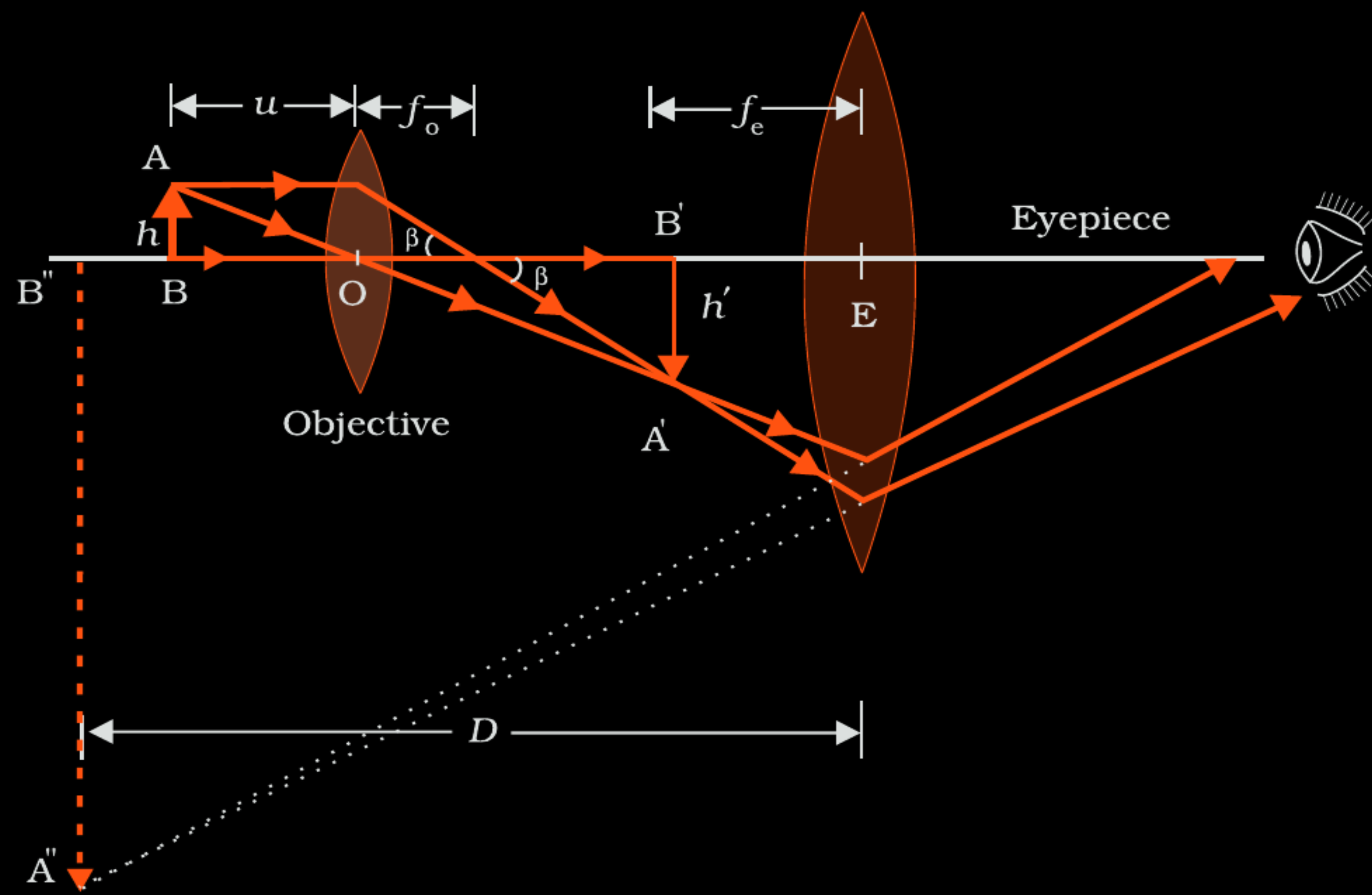
15. *Assertion (A) :* In a reflecting telescope, the image does not have chromatic aberration.

Reason (R) : Chromatic aberration occurs only due to refraction of light through an optical medium.

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(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A)

- 33.** (a) (i) Draw a ray diagram to show the image formation by a compound microscope. Obtain the expression for the total magnification of the microscope when the final image is formed at infinity.
- (ii) In a compound microscope, an object is placed at a distance of 1.5 cm from the objective of focal length 1.25 cm. The eyepiece has a focal length of 5 cm. The final image is formed at infinity. Calculate the distance between the objective and the eyepiece.



Note: Deduct ½ mark for not showing arrows with the rays.

Magnification produced by objective

$$m_0 = \frac{h'}{h} = \frac{L}{f_o}$$

Magnification produced by eye-piece

$$m_e = 1 + \frac{D}{f_e}$$

If the final image is formed at infinity

$$m_e = \frac{D}{f_e}$$

Total magnification

$$m = m_0 \times m_e$$

$$= \left(\frac{L}{f_o} \right) \left(\frac{D}{f_e} \right)$$

$$(i) \quad \frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$$

$$\frac{1}{v_o} - \frac{1}{(-1.5)} = \frac{1}{1.25}$$

$$v_o = 7.5 \text{ cm}$$

$$L = |v_o| + |f_e| \text{ as final image is formed at infinity } (v_e = \infty, u_e = f_e)$$

$$L = 7.5 + 5$$

$$L = 12.5 \text{ cm}$$

1½

½

½

½

½

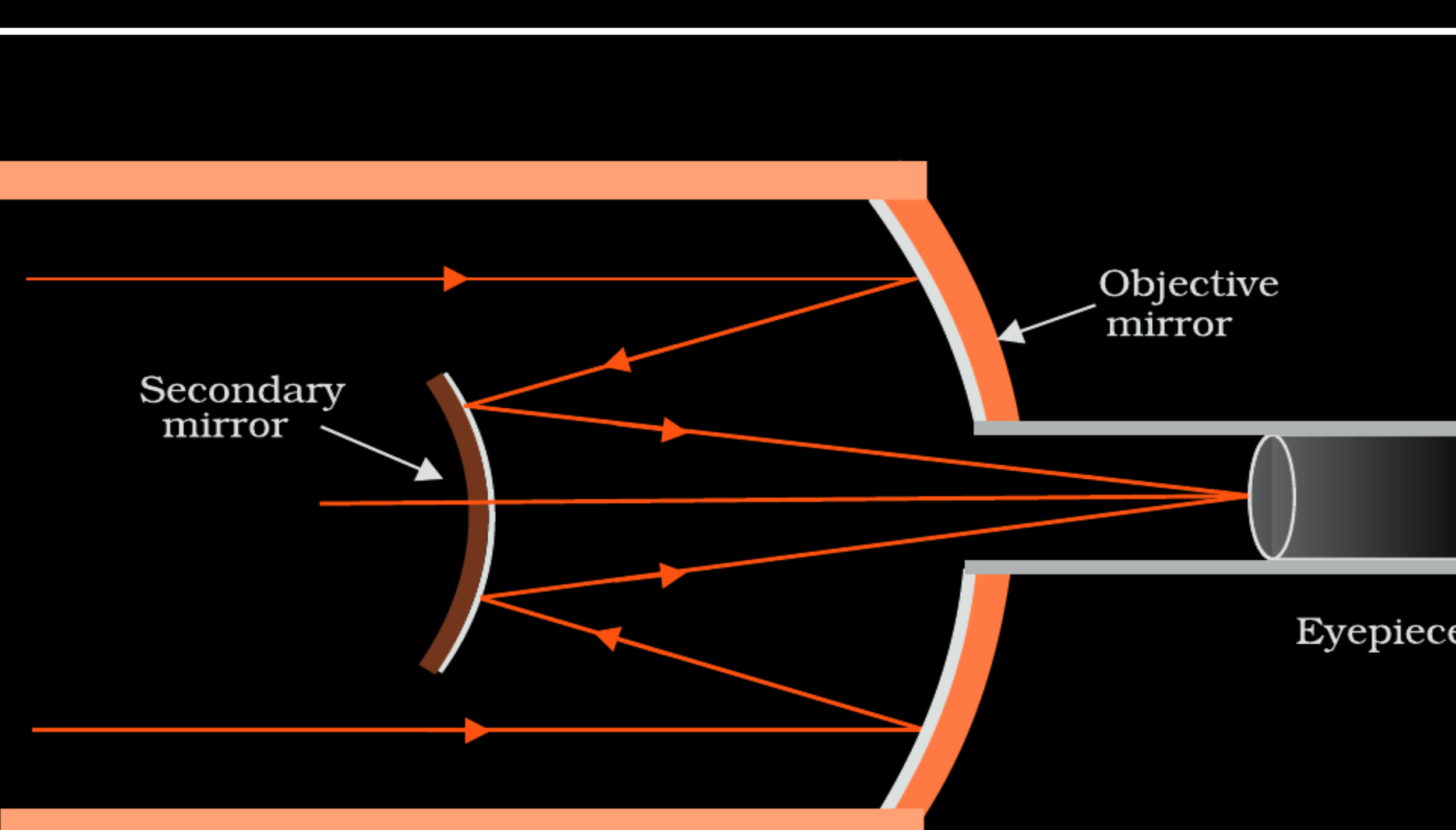
½

½

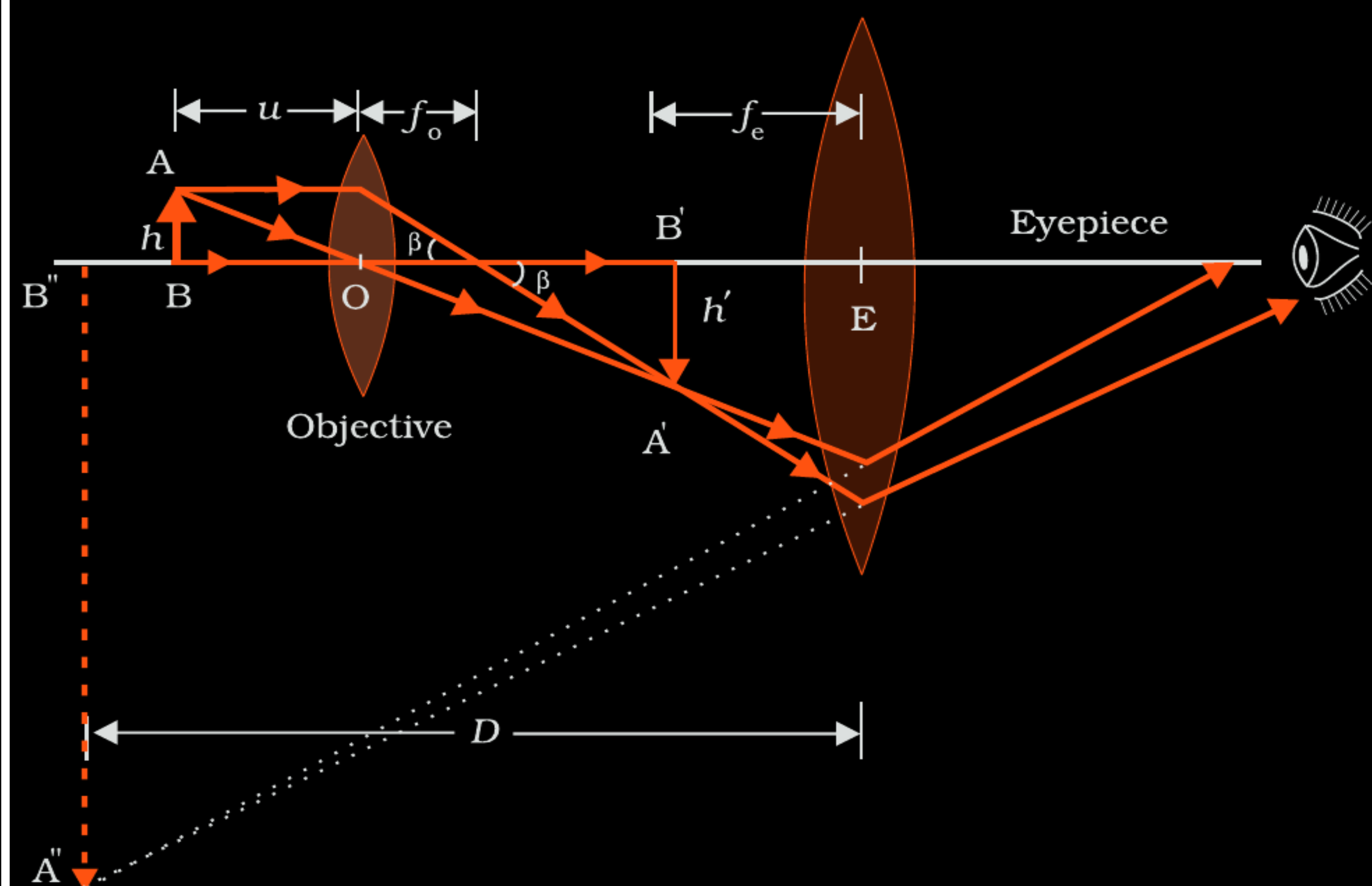
½

- 31.** (a) (i) Draw a ray diagram of a reflecting telescope (Cassegrain) and explain the formation of image. State two important advantages that a reflecting telescope has over a refracting telescope.
- (ii) In a refracting telescope, the focal length of the objective is 50 times the focal length of the eyepiece. When the final image is formed at infinity, the length of the tube is 102 cm. Find the focal lengths of the two lenses.

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	The parallel rays from a distant object are reflected by a large concave mirror. These rays are then reflected by a convex mirror placed just before the focus of concave mirror and are converged to a point outside the hole. The final image is viewed through eye piece. Advantages (any two) 1) No chromatic aberration. 2) Less spherical aberration 3) Less mechanical support required 4) Brighter Image 5) High resolving power. 6) High magnifying power	1 1
	(i) For image at infinity $ f_0 + f_e = L$ According to question $f_0 = 50 \times f_e$ $f_e + 50f_e = 102$ $f_e = 2 \text{ cm}$ $f_0 = 100 \text{ cm}$	$\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

- (b) (i) Write any two advantages of a compound microscope over a simple microscope. Draw a ray diagram for the image formation at the near point by a compound microscope and explain it.
- (ii) A thin planoconcave lens with its curved face of radius of curvature R is made of glass of refractive index n_1 . It is placed coaxially in contact with a thin equiconvex lens of same radius of curvature of refractive index n_2 . Obtain the power of the combination lens.



- (i) Advantages (any two)
- 1) Larger magnification
 - 2) Brighter image
- Any other valid advantage

$\frac{1}{2} + \frac{1}{2}$

(deduct $\frac{1}{2}$ mark for not showing arrow for ray diagram)

The lens nearest the object, called the objective, forms a real, inverted, magnified image of the object. This serves as the object for the second lens, the eye piece, functions like a simple microscope and produces final image which is enlarged and virtual.

1

(ii) Power of plano concave lens = $P_1 = \frac{-(n_1-1)}{R}$

$\frac{1}{2}$

Power of convex lens = $P_2 = (n_2-1) \left(\frac{2}{R} \right)$

$\frac{1}{2}$

$$P = P_1 + P_2$$

$\frac{1}{2}$

$$= \frac{(2n_2 - n_1 - 1)}{R}$$

$\frac{1}{2}$

32. (a) (i) What are the two main considerations for designing the objective and eyepiece lenses of an astronomical telescope ? Obtain the expression for magnifying power of the telescope when the final image is formed at infinity.

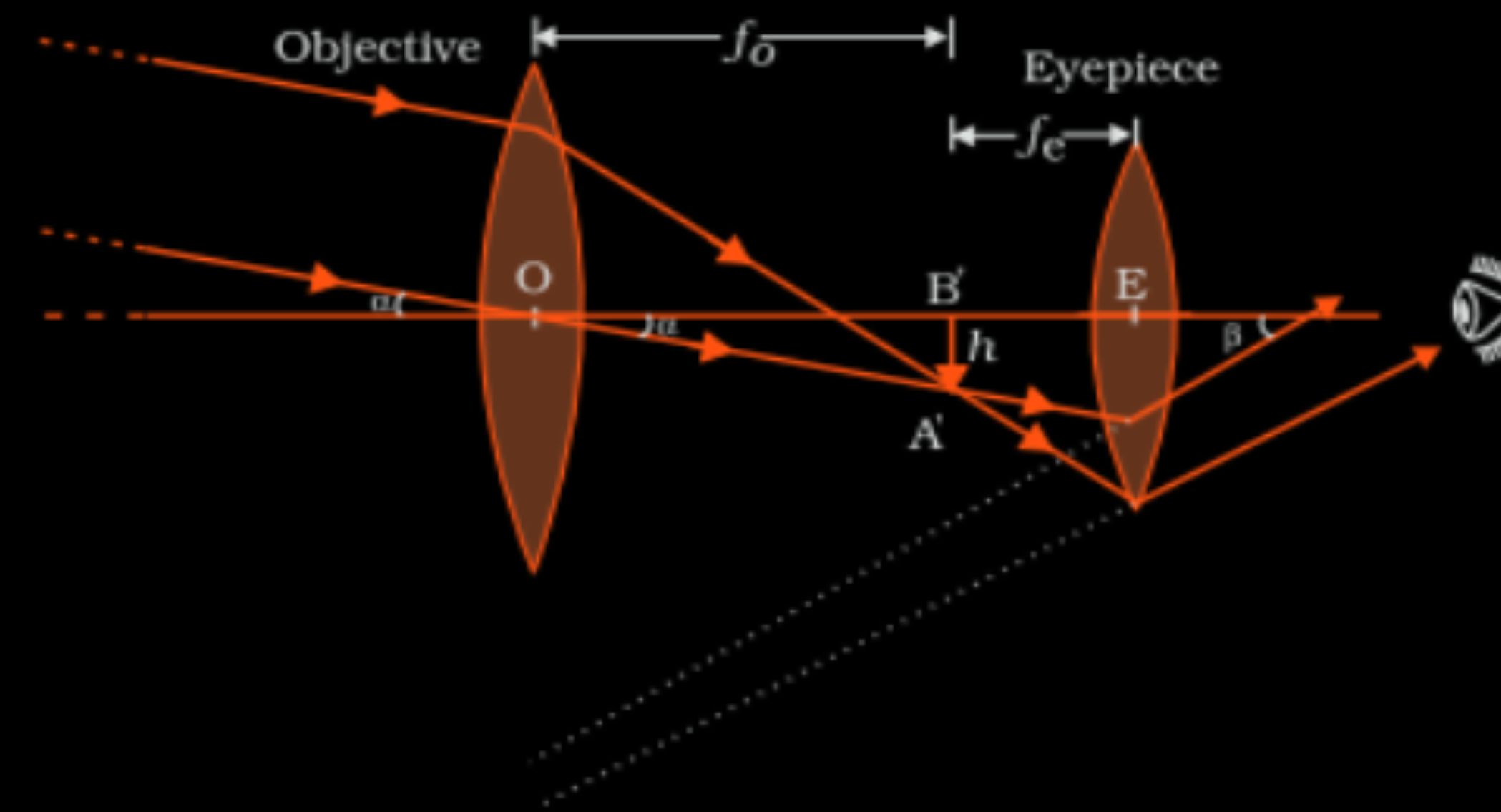
Two main considerations

Objective should have

1. Larger diameter
2. Larger focal length

Eye piece should have

1. Smaller diameter
2. Smaller focal length



Magnifying power of telescope

Magnifying power is the ratio of the angle β subtended at the eye by the final images to the angle α which the object subtends at the lens or eye

$$m \approx \frac{\beta}{\alpha} \approx \frac{h}{f_e} \cdot \frac{f_o}{h} = \frac{f_o}{f_e}$$

11. The focal lengths of the objective and the eyepiece of a compound microscope are 1 cm and 2 cm respectively. If the tube length of the microscope is 10 cm, the magnification obtained by the microscope for most suitable viewing by relaxed eye is :

(A) 250

(B) 200

(C) 150

(D) 125

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11. The focal lengths of the objective and the eyepiece of a compound microscope are 1 cm and 2 cm respectively. If the tube length of the microscope is 10 cm, the magnification obtained by the microscope for most suitable viewing by relaxed eye is :

(A) 250

(B) 200

(C) 150

(D) 125

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(D) 125

19. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. Calculate its magnifying power in normal adjustment and the distance of the image formed by the objective.

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19. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. Calculate its magnifying power in normal adjustment and the distance of the image formed by the objective.

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$$|m| = \frac{f_o}{f_e}$$
$$= \frac{150}{5} = 30$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{150} = \frac{1}{v} - \frac{1}{\infty}$$

$$v = 150 \text{ cm}$$

(Note: Award full credit of this part, if a student writes correct distance of image without calculation i.e. using object position at infinity.)

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- (b) In normal adjustment, for a refracting telescope, the distance between objective and eye piece lens is 1.00 m. If the magnifying power of the telescope is 19, find the focal length of the objective and the eyepiece lens.

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- (b) In normal adjustment, for a refracting telescope, the distance between objective and eye piece lens is 1.00 m. If the magnifying power of the telescope is 19, find the focal length of the objective and the eyepiece lens.

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2

Distance between objective and eyepiece

$$f_o + f_e = 1.00 \text{ m} = 100 \text{ cm}$$

Magnifying power

$$|m| = \frac{f_o}{f_e} = 19$$

On solving

$$f_o = 95 \text{ cm} = 0.95 \text{ m}$$

$$f_e = 5 \text{ cm} = 0.05 \text{ m}$$

$\frac{1}{2}$

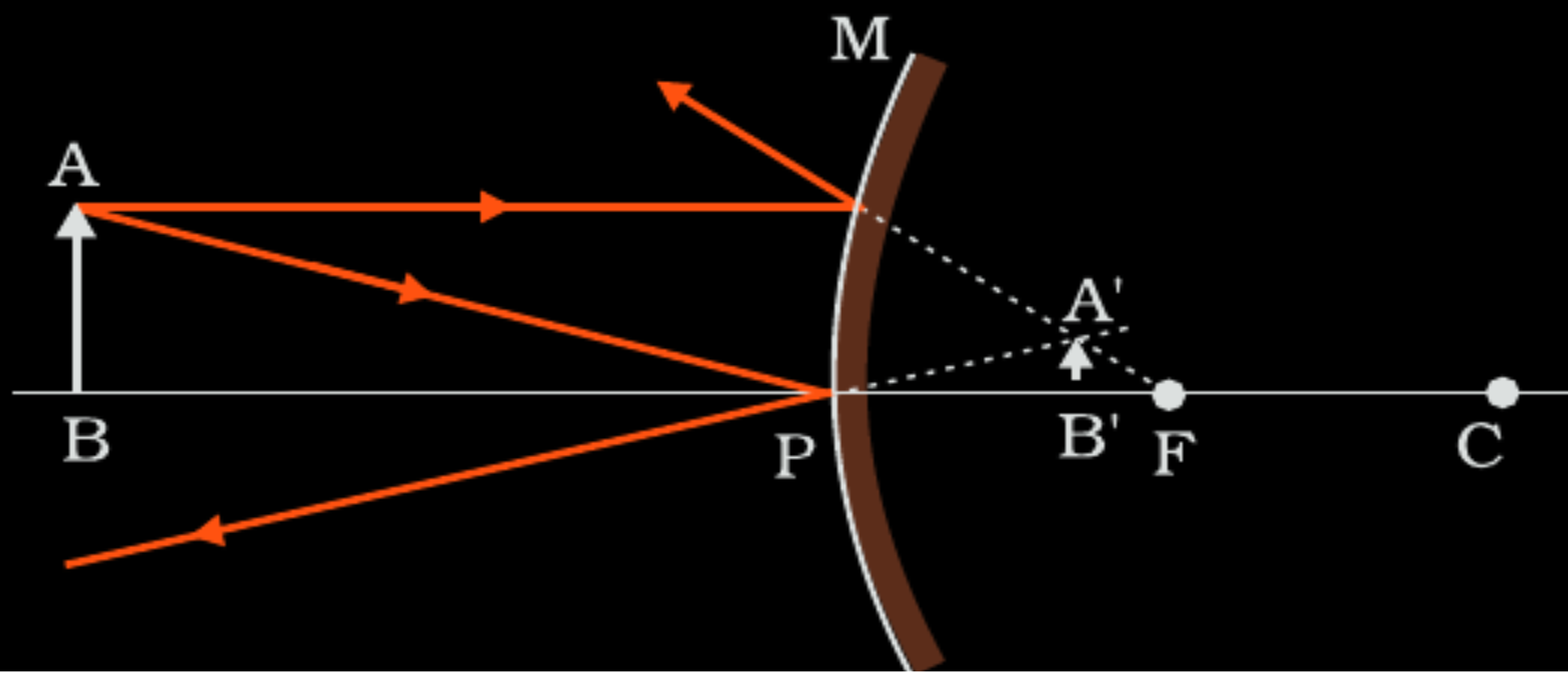
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

32. (a) (i) Draw a ray diagram for the formation of the image of an object by a convex mirror. Hence, obtain the mirror equation.
- (ii) Why are multi-component lenses used for both the objective and the eyepiece in optical instruments ?
- (iii) The magnification of a small object produced by a compound microscope is 200. The focal length of the eyepiece is 2 cm and the final image is formed at infinity. Find the magnification produced by the objective.

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For paraxial rays MP can be considered to be a straight line perpendicular to CP,
Therefore right angled triangles $A'B'F$ and MPF are similar

$$\frac{B'A'}{PM} = \frac{B'F}{FP}$$

$$\text{Or } \frac{B'A'}{BA} = \frac{B'F}{FP} \quad (\because PM = AB) \quad \text{-----(1)}$$

Since $\angle APB = \angle A'PB'$, the right angled triangles $A'PB'$ and AB

- 18.** The magnifying power of an astronomical telescope is 24. In normal adjustment, distance between its two lenses is 150 cm. Find the focal length of the objective lens.

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18. The magnifying power of an astronomical telescope is 24. In normal adjustment, distance between its two lenses is 150 cm. Find the focal length of the objective lens.

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2

Magnifying power = 24 , Distance between lenses = 150 cm

$$\frac{f_o}{f_e} = 24$$

$$f_o + f_e = 150 \text{ cm}$$

$$f_e = 6 \text{ cm}$$

$$f_o = 144 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

</

19. Explain the following :

2

- (a) For a simple microscope, the angular size of the object equals the angular size of the image. Yet it offers magnification.
- (b) Both plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ?

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- (a) For a simple microscope, the angular size of the object equals the angular size of the image. Yet it offers magnification.
- (b) Both plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ?

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(a) Yes, it offers magnification.

 $\frac{1}{2}$

We can keep the small object much closer to the eye than 25 cm and hence have it subtend a large angle.

 $\frac{1}{2}$

(b) Yes,

 $\frac{1}{2}$

Rays converging to a point behind a plane or convex mirror are reflected to a point in front of the mirror on a screen

14. *Assertion (A) :* The magnifying power of a compound microscope is negative.

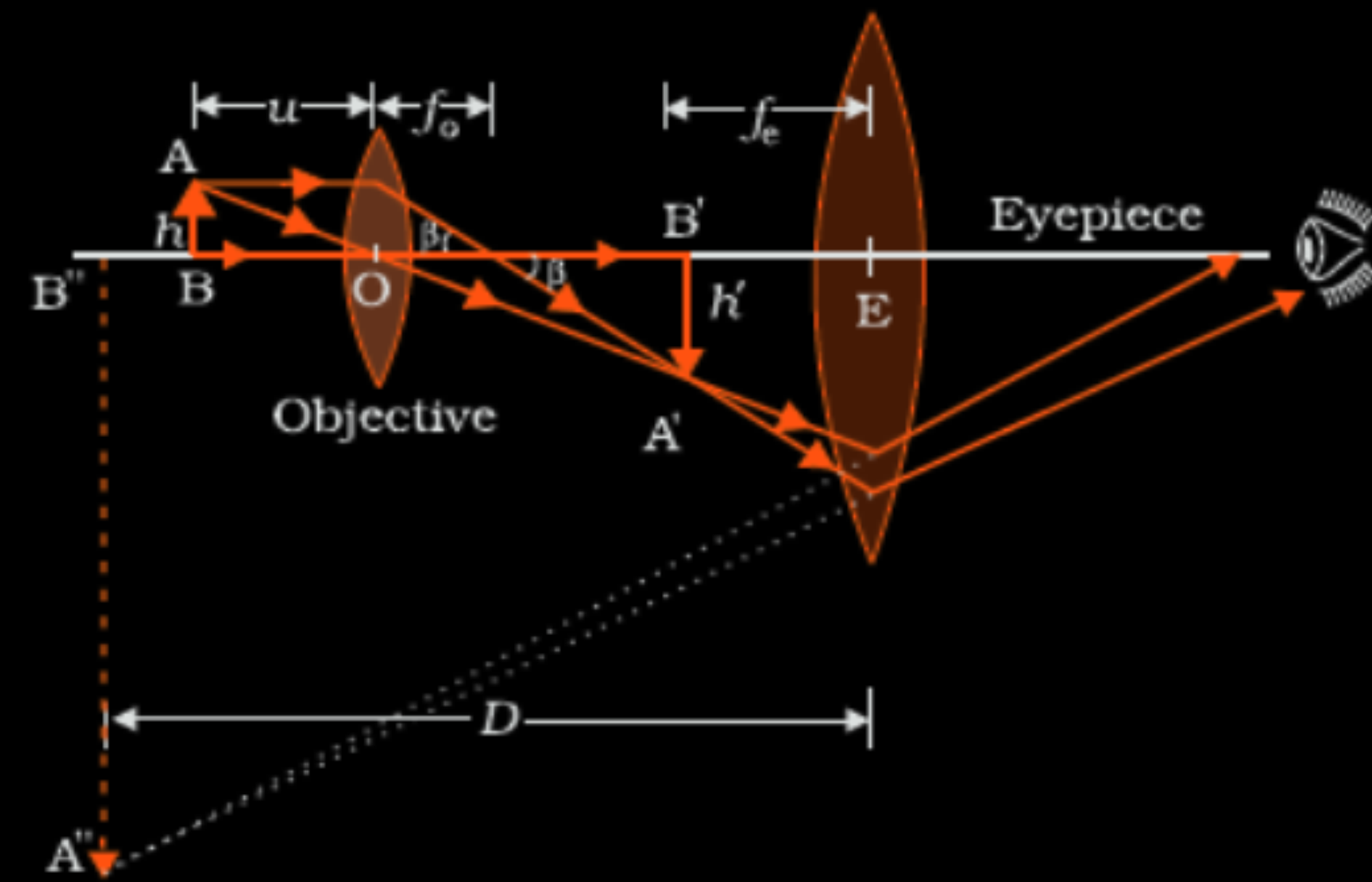
Reason (R) : The final image formed is erect with respect to the object.

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- (b) (i) Draw a labelled ray diagram of a compound microscope showing image formation at least distance of distinct vision. Derive an expression for its magnifying power.
- (ii) A telescope consists of two lenses of focal length 100 cm and 5 cm. Find the magnifying power when the final image is formed at infinity.

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(i)

**1**

The magnification obtained by eye-piece lens $m_e = \left(1 + \frac{D}{f_e}\right)$

 $\frac{1}{2}$

The magnification obtained by objective lens $m_o = \frac{v_0}{-u_0}$

 $\frac{1}{2}$

Hence the total magnifying power is

$$m = m_o \times m_e$$

 $\frac{1}{2}$

$$= \frac{v_0}{-u_0} \left(1 + \frac{D}{f_e}\right)$$

 $\frac{1}{2}$

$$(ii) m = \left| \frac{f_o}{f_e} \right|$$

1

Identification of focal length of objective and eyepiece

$$f_o = 100cm$$

 $\frac{1}{2}$

$$f_e = 5cm$$

$$m = \left| \frac{100}{5} \right| = 20$$

 $\frac{1}{2}$

(i) (1) To improve image quality by minimizing various optical aberrations in lenses. (2) To achieve large magnification Alternatively: Magnifying Power $m = \left(\frac{L}{f_o} \right) \left(\frac{D}{f_e} \right)$ (3) Because of this position our eye will collect all the light refracted by the objective which results in relaxation of the eye. Alternatively: To increase the field of view & for relaxation of eye.	1 1 1
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- (b) (i) Draw a ray diagram showing the image formation by a compound microscope. Hence obtain the expression for total magnification when the image is formed at least distance of distinct vision.
- (ii) A compound microscope consists of an objective lens of focal length 2.0 cm and an eyepiece of focal length 6.0 cm. If they are separated by a distance of 24 cm, find the total magnification when the image is formed at infinity.

5

Magnification due to objective

$$m_0 = \frac{h'}{h} = \frac{L}{f_0} \quad \left(\because \tan \beta = \frac{h}{f_0} = \frac{h'}{L} \right)$$

Magnification due to eye piece when final image is formed at the near point

$$m_e = 1 + \frac{D}{f_e}$$

Total magnification m

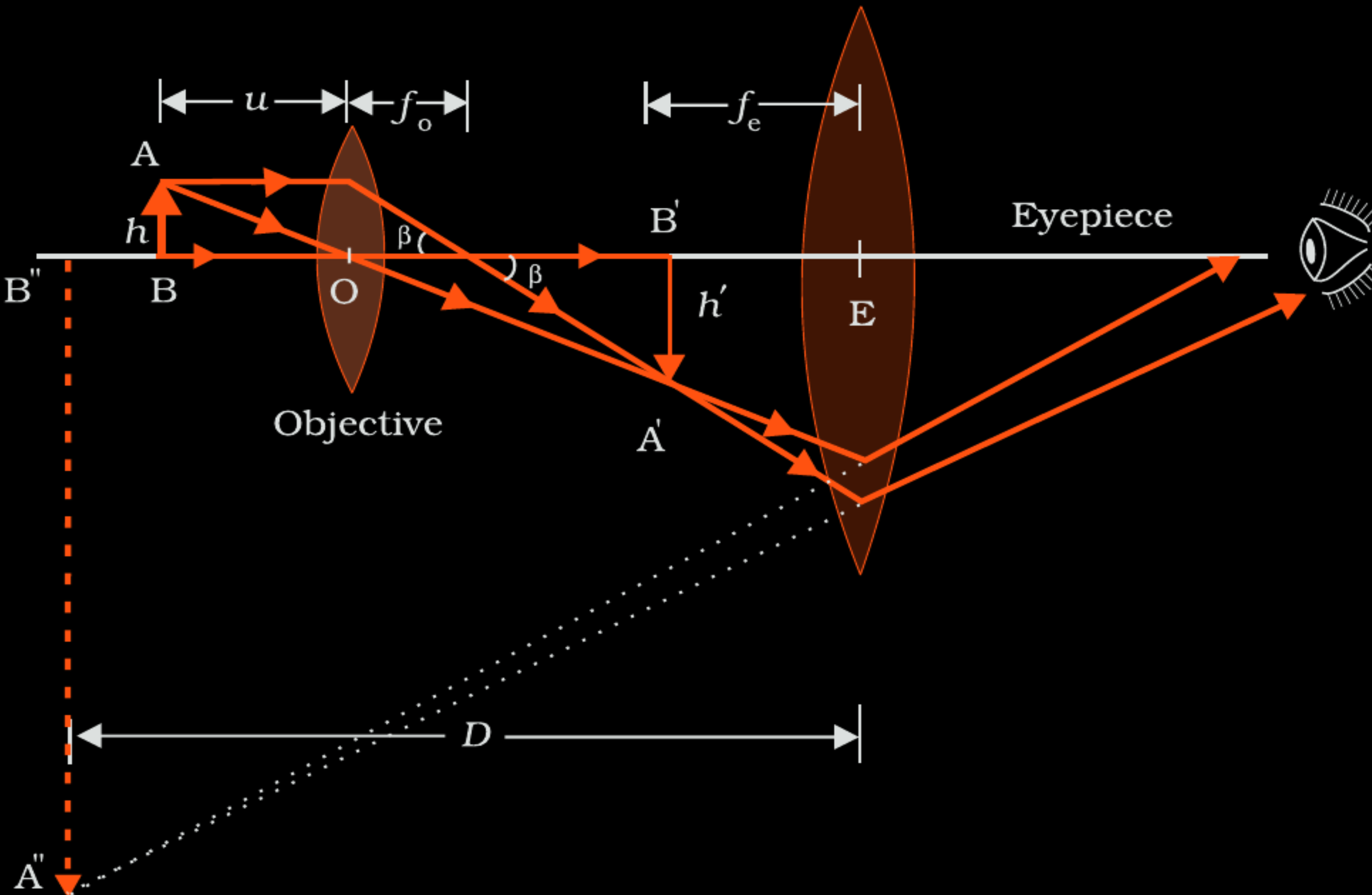
$$m = m_0 m_e$$

$$m = \frac{L}{f_0} \left(1 + \frac{D}{f_e} \right)$$

$$m = \frac{L}{f_0} \left(1 + \frac{D}{f_e} \right)$$

$$(ii) \quad m = \frac{LD}{f_0 f_e}$$

$$m = \frac{24 \times 25}{2 \times 6} = \frac{600}{12} = 50$$



$\frac{1}{2}$

$\frac{1}{2}$

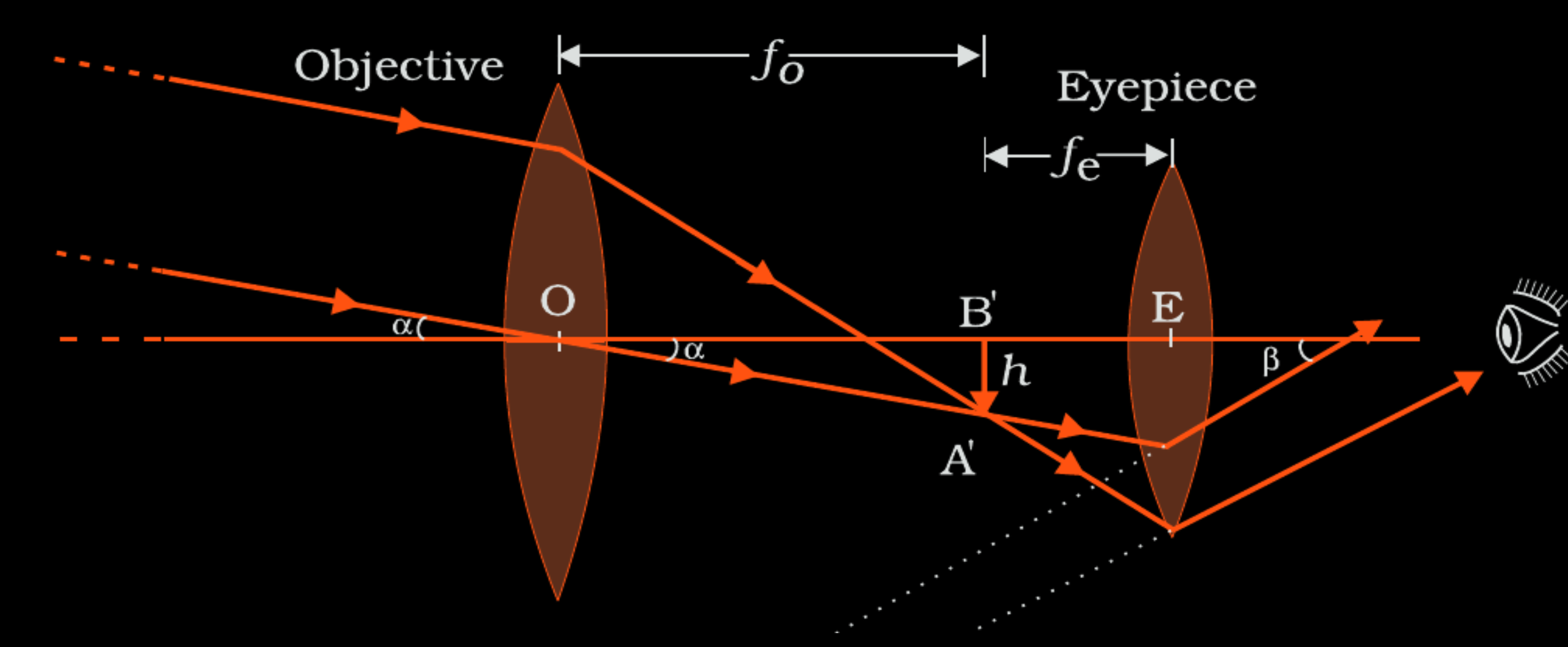
$\frac{1}{2}$

$\frac{1}{2}$

1

1

- 31.** (a) (i) Draw a labelled ray diagram of an astronomical telescope to show the image formation of a distant object by it in normal adjustment. What are the main considerations required in selecting the objective and eyepiece lenses so that the telescope has large magnifying power and high resolution ?
- (ii) A biconvex lens of focal length 20 cm is immersed in water, whose refractive index is $\frac{4}{3}$. Find the change, if any, in the nature and the focal length of the lens. Refractive index of the material of convex lens is $\frac{3}{2}$.



For large magnifying power

$$f_o > f_e$$

For high resolution

Aperture of objective should be large

$$(ii) \frac{1}{f_a} = (n_g - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\frac{1}{f_a} = \left(\frac{n_g}{n_w} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\frac{f_w}{f_a} = \frac{(n_g - 1)}{\left(\frac{n_g}{n_w} - 1 \right)}$$

$$\frac{f_w}{20} = \frac{\left(\frac{3}{2} - 1 \right)}{\left(\frac{3}{2} \times \frac{3}{4} - 1 \right)} = \frac{\frac{1}{2}}{\frac{1}{8}} = 4$$

$$f_w = 80 \text{ cm}$$

$$\begin{aligned} \text{Changes in focal length } \Delta f &= 80 - 60 \\ &= 20 \text{ cm} \end{aligned}$$

Nature of the lense is converging because f_w is positive

1/2

1/2

1/2

1/2

1/2

1/2

1/2

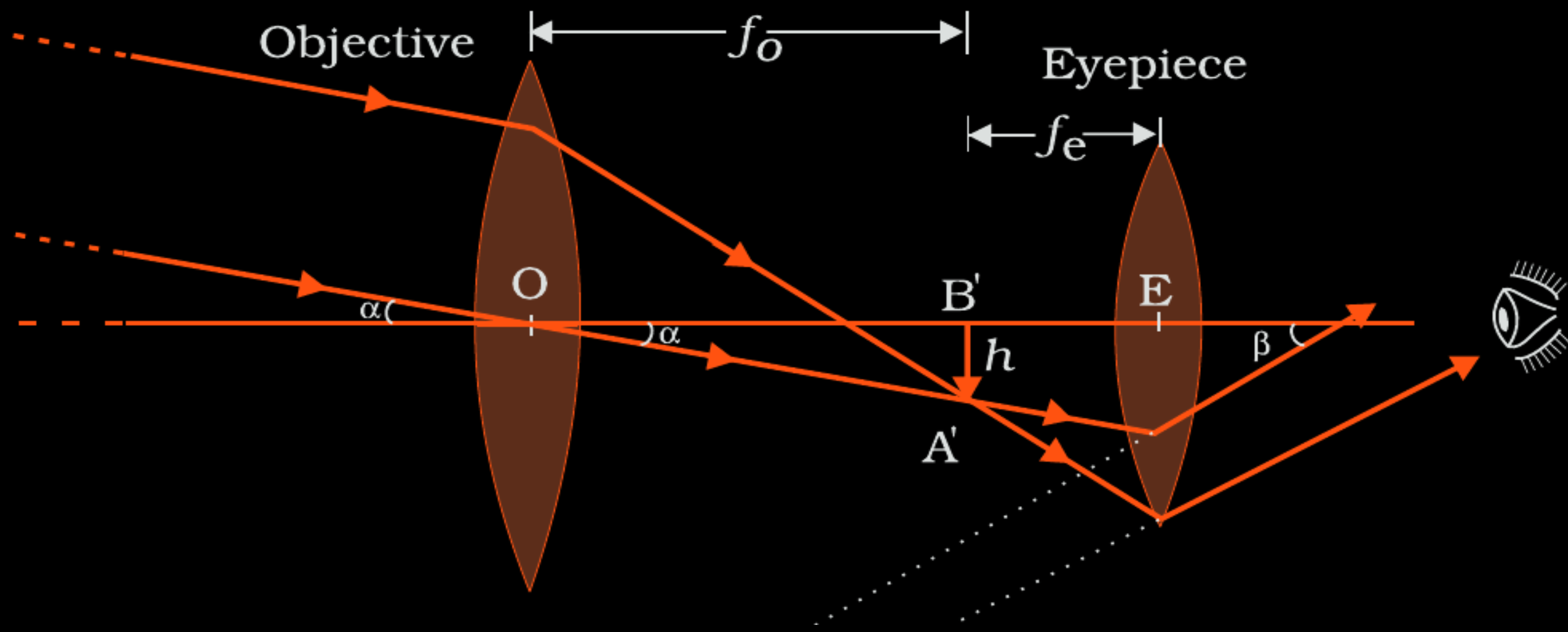
1/2

- (i) Answer the following giving reasons :
- (1) The angular size of the image equals the angular size of the object in a simple microscope, yet it offers magnification.
 - (2) Both the objective and the eyepiece of a compound microscope have short focal lengths.
 - (3) A microscope and a telescope play different roles with respect to resolution and magnification of the image.

(i) 1. Because the image is at minimum distance of distinct vision(D) and object is very close to the eye at distance d ($D > d$) Angular magnification $\theta = \frac{h_2}{h_1} = \frac{D}{d} \Rightarrow h_2 > h_1$ h_1 -size of object h_2 -size of image	1
2. Magnification of a compound microscope $m = \left(\frac{L}{f_0}\right)\left(\frac{D}{f_e}\right)$	$\frac{1}{2}$
To achieve large magnification (m), f_0 and f_e must be short.	$\frac{1}{2}$
3. Microscope - magnifies	$\frac{1}{2}$
Telescope - Resolves	$\frac{1}{2}$

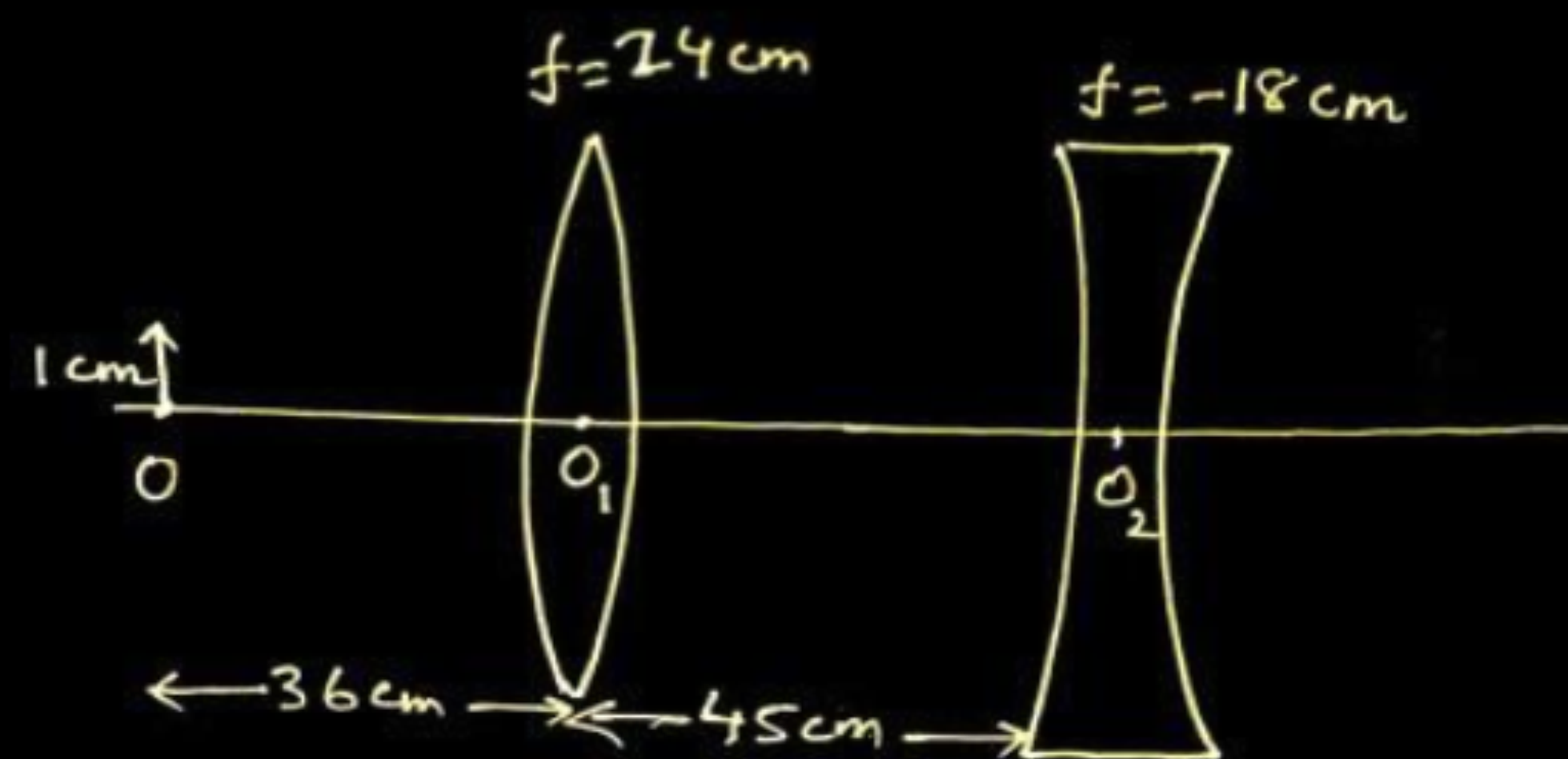
- 33.** (a) (i) Draw a ray diagram to show how the final image is formed at infinity in an astronomical refracting telescope. Obtain an expression for its magnifying power.
- (ii) Two thin lenses L_1 and L_2 , L_1 being a convex lens of focal length 24 cm and L_2 a concave lens of focal length 18 cm are placed coaxially at a separation of 45 cm. A 1 cm tall object is placed in front of the lens L_1 at a distance of 36 cm. Find the location and height of the image formed by the combination.

5



From the diagram $\beta = \frac{h}{f_e}$
 and $\alpha = \frac{h}{f_o}$
 Magnifying Power = $\frac{f_o}{f_e}$

(ii)



1 1/2

1/2

1/2

1/2

For lens L₁,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-36} = \frac{1}{24}$$

$$\frac{1}{v} = \frac{1}{24} - \frac{1}{36}$$

$$\frac{1}{v} = \frac{3-2}{72} = \frac{1}{72}$$

$$v = 72 \text{ cm}$$

For lens L₂:

$$\frac{1}{v'} - \frac{1}{u'} = \frac{1}{f'}$$

$$\frac{1}{v'} - \frac{1}{(72-45)} = \frac{1}{-18}$$

$$\frac{1}{v'} = \frac{1}{-18} + \frac{1}{27}$$

$$\frac{1}{v'} = \frac{-3+2}{54} = \frac{-1}{54}$$

$$v' = -54 \text{ cm}$$

Final distance $v'_1 = -54 - (-45)$

$v'_1 = -9 \text{ cm}$ (to the left of convex lens)

Magnification $\frac{h_i}{h_o} = \frac{v'_1}{u}$

$$\frac{h_i}{1} = \frac{-9}{-36} \Rightarrow h_i = +\frac{1}{4} \text{ cm}$$

1/2

1/2

1/2

1/2

(b) Why is a reflecting telescope preferred over a refracting telescope ?

Justify your answer giving two reasons.

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2

(b) Why is a reflecting telescope preferred over a refracting telescope ?

Justify your answer giving two reasons.

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2

1 No Chromatic Aberration - No refraction in mirrors 2 No Spherical Aberration - Due to use of parabolic reflector 3 Easy mechanical support required - Mirrors weigh less and can be supported over entire back surface. 4 High resolving power – Due to Mirror with large diameter are better. 5 Brighter image – Large mirrors gather more light waves. (Any two)	1+1
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- (b) (i) Draw a labelled ray diagram showing the image formation by a refracting telescope. Define its magnifying power. Write two limitations of a refracting telescope over a reflecting telescope.
- (ii) The focal lengths of the objective and the eye-piece of a compound microscope are 1.0 cm and 2.5 cm respectively. Find the tube length of the microscope for obtaining a magnification of 300.

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(Note deduct ½ mark if a student does not show the direction of propagation of the light.)

Alternatively

Give full credit for ray diagram if a candidate draws ray diagram for final image at the near point.

Magnifying power of a telescope – It is defined as the ratio the angle subtended at the eye by the final image to the angle subtended by the object at the lens or the eye.

Two limitations of a refracting telescope over a reflecting telescope.

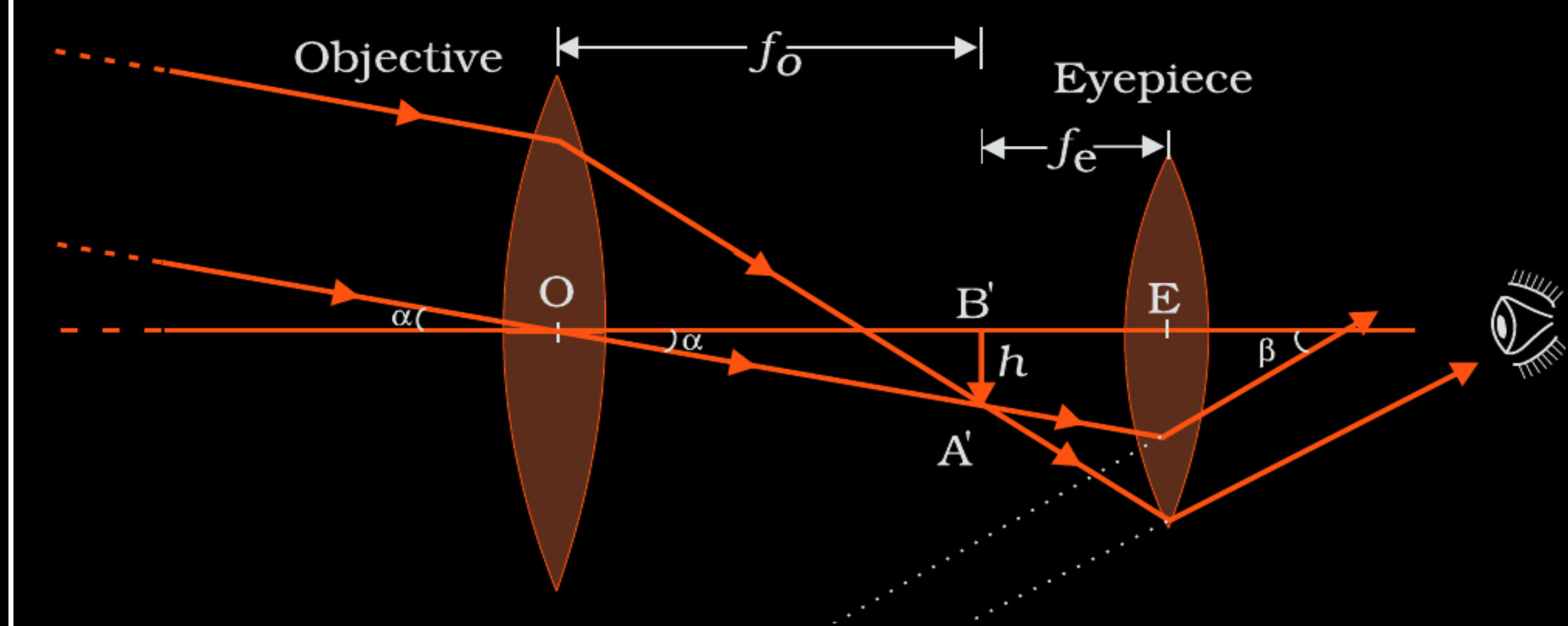
- (i) Less resolving power.
 - (ii) Difficult mechanical support.
 - (iii) Less bright image.
 - (iv) Suffers chromatic aberration.
 - (v) Image suffers with spherical aberration.
- (Any two of the above)

$$f_o = 1.0 \text{ cm} , f_e = 2.5 \text{ cm} , m = 300 , D = 25 \text{ cm} , L = ?$$

$$|m| = \frac{L}{f_o} \cdot \frac{D}{f_e}$$

$$300 = \frac{L}{1.0} \cdot \frac{25}{2.5}$$

$$L = 30 \text{ cm}$$



1

½ + ½

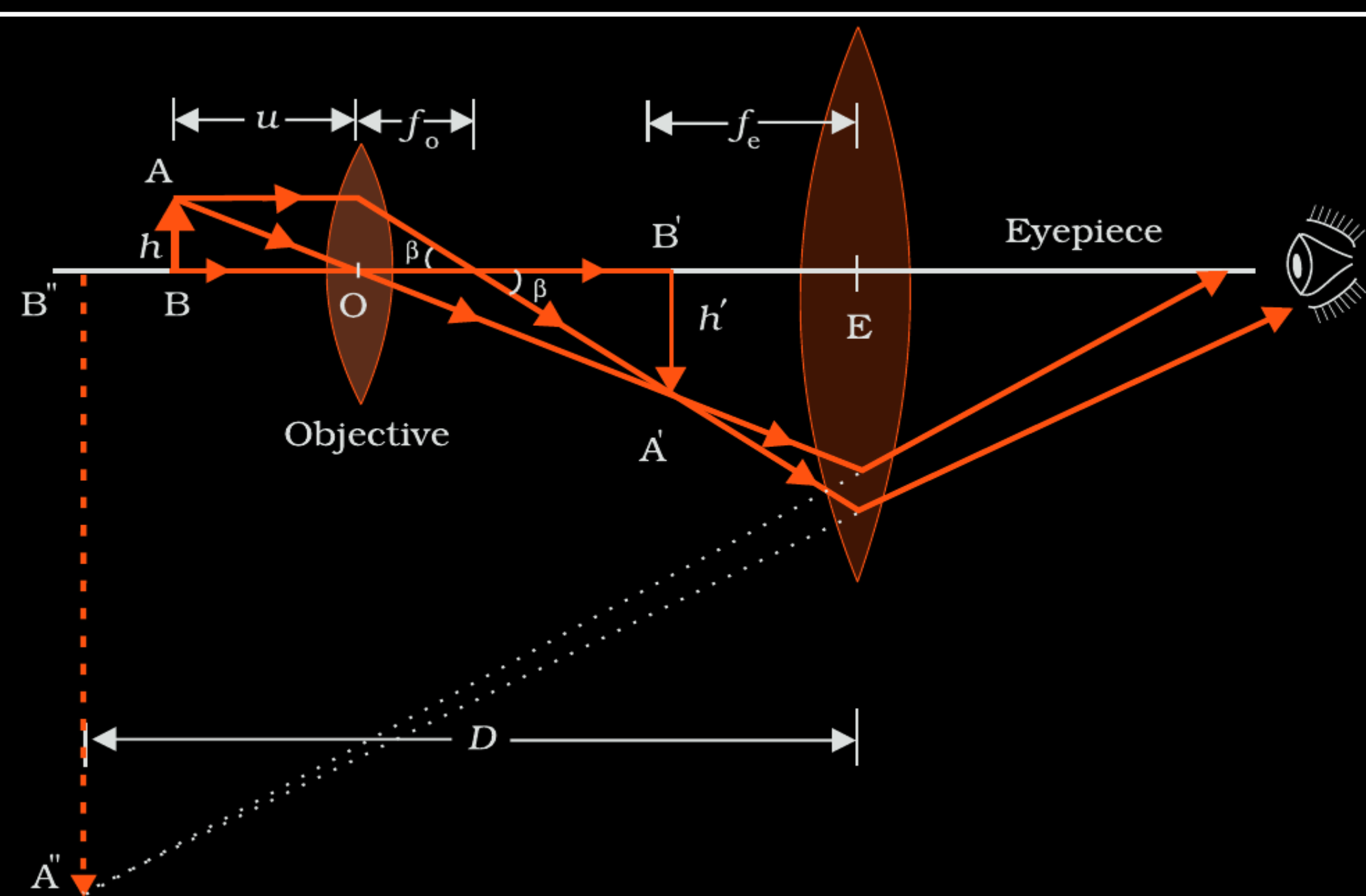
½

½

½

32. (a) (i) Draw a ray diagram to show the working of a compound microscope. Obtain the expression for the total magnification for the final image to be formed at the near point. **3**
- (ii) In a compound microscope an object is placed at a distance of 1.5 cm from the objective of focal length 1.25 cm. If the eye-piece has a focal length of 5 cm and the final image is formed at the near point, find the magnifying power of the microscope. **2**

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1½

Linear magnification due to the objective is

$$\tan \beta = \left(\frac{h}{f_0} \right) = \left(\frac{h'}{L} \right)$$

$$m_0 = \frac{h'}{h} = \frac{L}{f_0}$$

Here L is the distance between the second focal point of the objective and the first focal point of the eyepiece.

Linear magnification due to eyepiece is

$$m_e = \left(1 + \frac{D}{f_e} \right)$$

Thus, the total magnification is, $m = m_0 \times m_e$

$$m = \frac{L}{f_0} \left(1 + \frac{D}{f_e} \right)$$

Note: full credit of the derivation should be given if a student derives

$$m = \frac{v_0}{u_0} \left(1 + \frac{D}{f_e} \right)$$

½

½

½

$$u_0 = -1.5 \text{ cm}$$

$$f_0 = 1.25 \text{ cm}$$

$$f_e = 5 \text{ cm}$$

$$D = 25 \text{ cm}$$

$$\frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$$

$$\frac{1}{v_0} = \frac{1}{1.25} - \frac{1}{1.5} = \frac{2}{15}$$

$$v_0 = \frac{15}{2} \text{ cm}$$

$$|m_0| = \frac{v_0}{u_0} = \frac{15}{2} \times \frac{1}{1.5} = 5$$

$$|m_e| = \left(1 + \frac{D}{f_e} \right)$$

$$|m_e| = 1 + \frac{25}{5} = 6$$

$$m = m_0 \times m_e = 5 \times 6 = 30$$

½

½

½

½

- (b) (i) Draw a ray diagram for the formation of image of an object by an astronomical telescope, in normal adjustment. Obtain the expression for its magnifying power.
- (ii) The magnifying power of an astronomical telescope in normal adjustment is 2.9 and the objective and the eyepiece are separated by a distance of 150 cm. Find the focal lengths of the two lenses.

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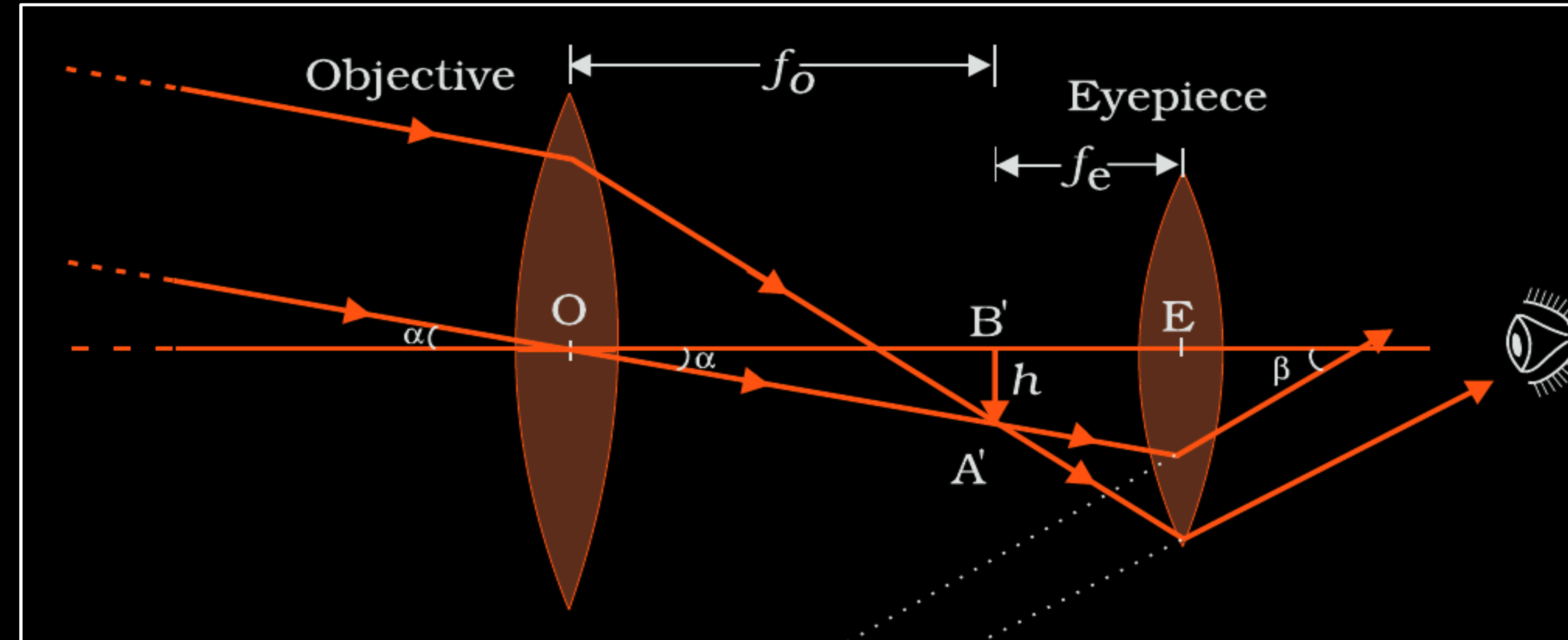
Magnifying power (m)

$$m \approx \frac{\beta}{\alpha}$$

$$\beta = \frac{h}{f_e}$$

$$\alpha = \frac{h}{f_o}$$

$$m = \frac{f_o}{f_e}$$



(ii) $m = 2.9$, $d = 150 \text{ cm}$ (Given)

$$m = \frac{f_o}{f_e} = \frac{29}{10}$$

$$f_o + f_e = 150$$

$$f_e = 38.5 \text{ cm}$$

$$f_o = 150 - 38.5 = 111.5 \text{ cm}$$

$1\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

8. (a) (i) Both the objective and the eyepiece of a compound microscope have short focal lengths. Explain.
- (ii) Two convex lenses of focal lengths 1.25 cm and 5.0 cm are used to design a compound microscope with an angular magnification of 30. Find the required tube length of the microscope for the final image to form at infinity.

8. (a) (i) Both the objective and the eyepiece of a compound microscope have short focal lengths. Explain.
- (ii) Two convex lenses of focal lengths 1.25 cm and 5.0 cm are used to design a compound microscope with an angular magnification of 30. Find the required tube length of the microscope for the final image to form at infinity.

CBSE 2022 C

3

(a) (i) The magnification produced by a compound microscope is inversely proportional to the focal length of the objective and the eyepiece. Hence short focal length increases magnification.

Alternatively

$m = \left(\frac{L}{f_0}\right) \left(\frac{D}{f_e}\right)$, where f_0 and f_e are small then magnification m will be large.

$$(ii) \quad m = m_0 \times m_e = \frac{L \times D}{f_0 \times f_e}$$

$$30 = \frac{L \times (25)}{1.25 \times 5}$$

$$\therefore L = 7.5 \text{ cm}$$

1

1

$\frac{1}{2}$

$\frac{1}{2}$

- 6.** (a) Draw a ray diagram to show the formation of an image at the least distance of distinct vision, by a compound microscope. Hence, obtain an expression for its angular magnification.

CBSE 2022 C

6. (a) Draw a ray diagram to show the formation of an image at the least distance of distinct vision, by a compound microscope. Hence, obtain an expression for its angular magnification.

CBSE 2022 C

3

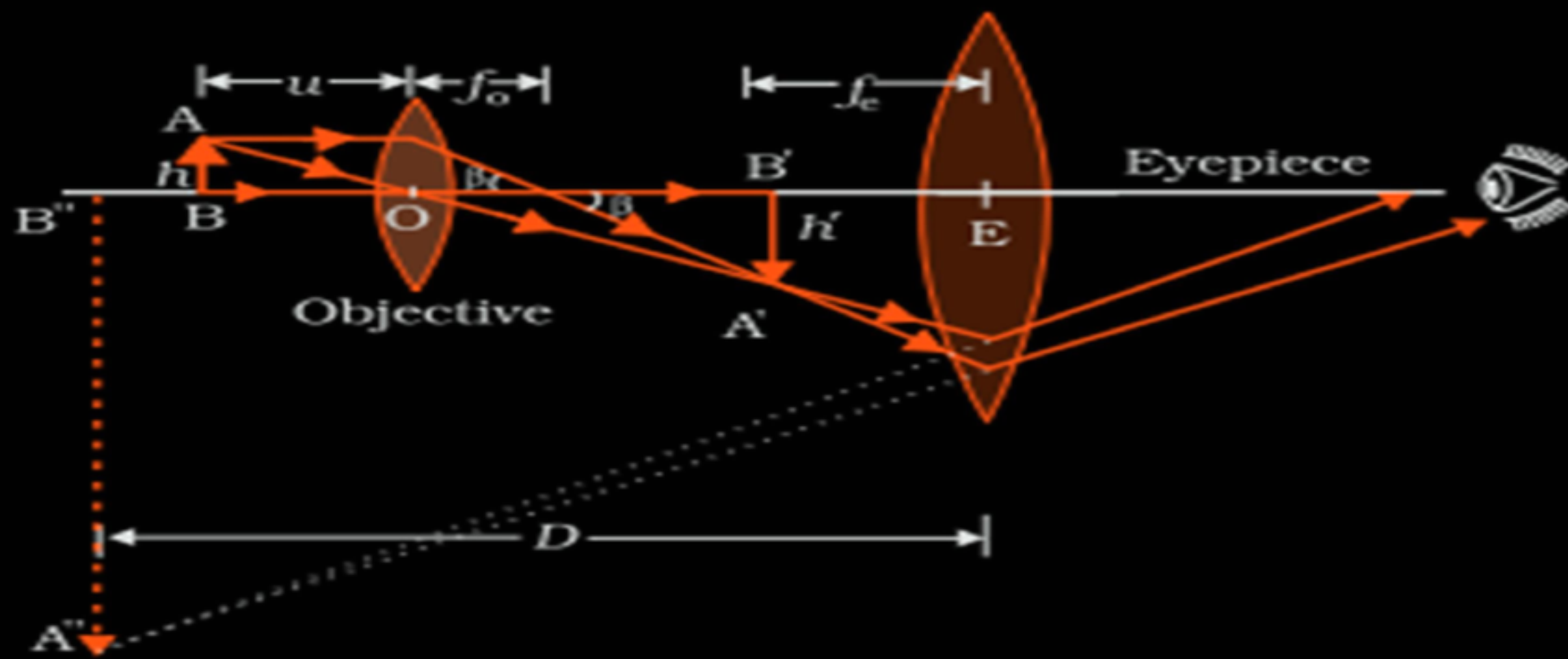
	Linear magnification due to objective (m_o) $m_o = \frac{h'}{h} = \frac{L}{f_o} \text{----- (1)}$ Linear magnification due to eye piece (m_e) when final image is formed at near point (D) $m_e = 1 + \frac{D}{f_e} \text{-----(2)}$ Total magnification $m = m_o \times m_e$ $m = \frac{L}{f_o} \left(1 + \frac{D}{f_e} \right)$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
--	--	---

6. With the help of a ray diagram, show how a compound microscope forms a magnified image of a tiny object, at least distance of distinct vision. Hence derive an expression for the magnification produced by it.

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6. With the help of a ray diagram, show how a compound microscope forms a magnified image of a tiny object, at least distance of distinct vision. Hence derive an expression for the magnification produced by it.

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(Note: Deduct $\frac{1}{2}$ mark if the arrows are not drawn in the ray diagram)

The linear magnification due to objective lens

1½

$$m_o = \frac{h'}{h} = \frac{L}{f_o} \text{ -----(i)}$$

h = size of object

h' = size of first image

$$\text{As } \tan \beta = \frac{h}{f_o} = \frac{h'}{L}$$

magnification due to eye piece

$$m_e = 1 + \frac{D}{f_e} \text{ -----(ii)}$$

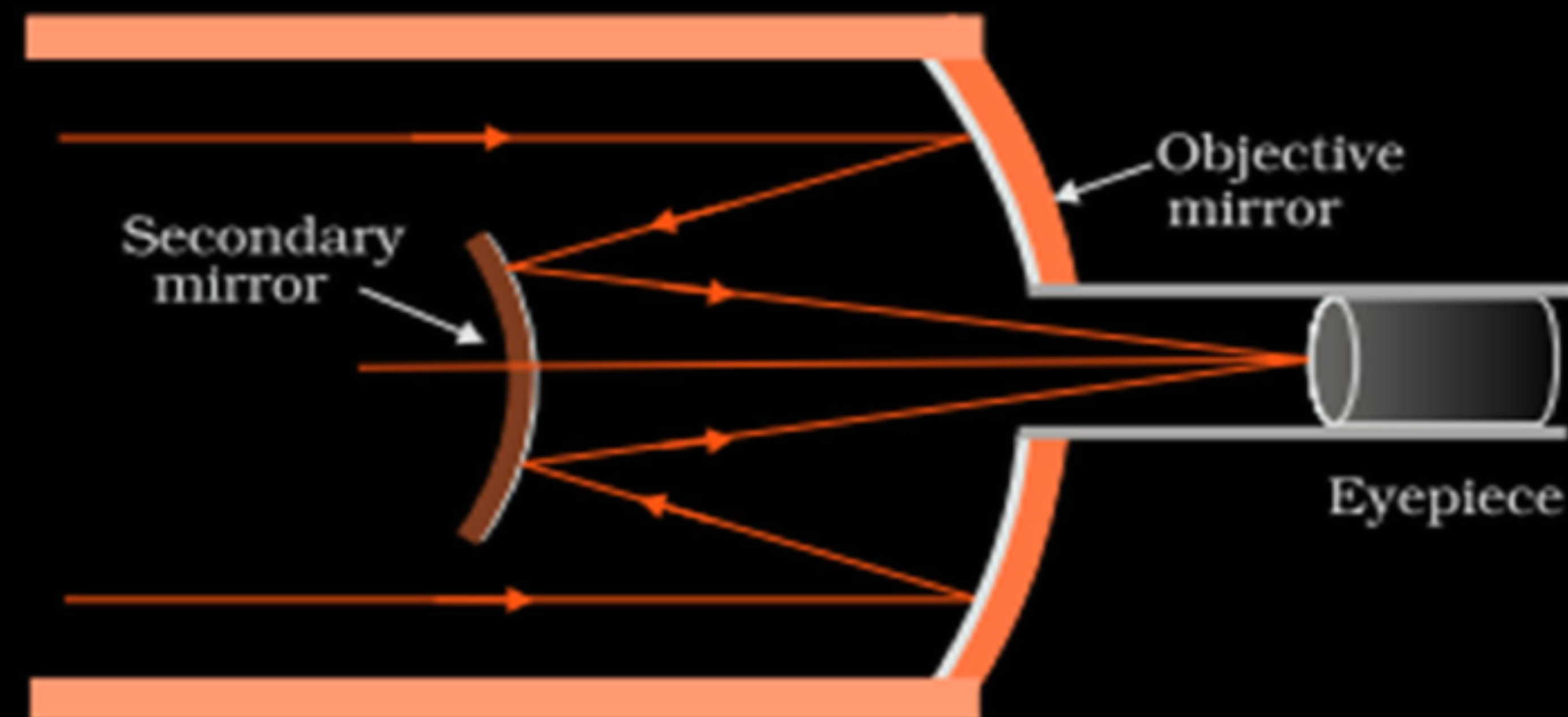
Total Magnification $m = m_o \times m_e$

$$m. = \frac{L}{f_o} \left(1 + \frac{D}{f_e} \right)$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

- (b) With the help of a ray diagram explain the working of a reflecting telescope. Mention two advantages of a reflecting telescope over a refracting telescope.

- (b) With the help of a ray diagram explain the working of a reflecting telescope. Mention two advantages of a reflecting telescope over a refracting telescope.



Working: Parallel beam of light gathered by objective mirror is reflected to the secondary mirror, which further forms the image in front of the eyepiece.

(**Note:** Deduct $\frac{1}{2}$ marks for not showing the direction of propagation of rays and give full credit for the ray diagram of Newtonian telescope)

Two Advantages (Any Two)

- High resolving power
- No chromatic aberration
- Reduced spherical aberration
- Brighter image is formed
- Easy mechanical support
- Large magnifying power

1 $\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

12. A compound microscope consists of two converging lenses. One of them, of smaller aperture and smaller focal length is called objective and the other of slightly larger aperture and slightly larger focal length is called eye-piece. Both the lenses are fitted in a tube with an arrangement to vary the distance between them. A tiny object is placed in front of the objective at a distance slightly greater than its focal length. The objective produces the image of the object which acts as an object for the eye-piece. The eye piece, in turn produces the final magnified image. $1 \times 5 = 5$

- I. In a compound microscope the images formed by the objective and the eye-piece are respectively
- | | |
|----------------------|-------------------|
| (A) virtual, real | (B) real, virtual |
| (C) virtual, virtual | (D) real, real |

12. A compound microscope consists of two converging lenses. One of them, of smaller aperture and smaller focal length is called objective and the other of slightly larger aperture and slightly larger focal length is called eye-piece. Both the lenses are fitted in a tube with an arrangement to vary the distance between them. A tiny object is placed in front of the objective at a distance slightly greater than its focal length. The objective produces the image of the object which acts as an object for the eye-piece. The eye piece, in turn produces the final magnified image. $1 \times 5 = 5$

I. In a compound microscope the images formed by the objective and the eye-piece are respectively

(A) virtual, real

(B) real, virtual

(C) virtual, virtual

(D) real, real

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(B) real , virtual

- II. The magnification due to a compound microscope *does not* depend upon
- (A) the aperture of the objective and the eye-piece
 - (B) the focal length of the objective and the eye-piece
 - (C) the length of the tube
 - (D) the colour of the light used

- II. The magnification due to a compound microscope *does not* depend upon
- (A) the aperture of the objective and the eye-piece
 - (B) the focal length of the objective and the eye-piece
 - (C) the length of the tube
 - (D) the colour of the light used

(A) The aperture of the objective and the eye piece

III. Which of the following is *not correct* in the context of a compound microscope ?

- (A) Both the lenses are of short focal lengths.
- (B) The magnifying power increases by decreasing the focal lengths of the two lenses.
- (C) The distance between the two lenses is more than $(f_o + f_e)$.
- (D) The microscope can be used as a telescope by interchanging the two lenses.

III. Which of the following is *not correct* in the context of a compound microscope ?

- (A) Both the lenses are of short focal lengths.
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(D) The microscope can be used as a telescope by interchanging the two lenses.

IV. A compound microscope consists of an objective of 10X and an eye-piece of 20X. The magnification due to the microscope would be

(A) 2

(B) 10

(C) 30

(D) 200

IV. A compound microscope consists of an objective of 10X and an eye-piece of 20X. The magnification due to the microscope would be

(A) 2

(B) 10

(C) 30

(D) 200

(D) 200

V. The focal lengths of objective and eye-piece of a compound microscope are 1.2 cm and 3.0 cm respectively. The object is placed at a distance of 1.25 cm from the objective. If the final image is formed at infinity, the magnifying power of the microscope would be

(A) 100

(B) 150

(C) 200

(D) 250

V. The focal lengths of objective and eye-piece of a compound microscope are 1.2 cm and 3.0 cm respectively. The object is placed at a distance of 1.25 cm from the objective. If the final image is formed at infinity, the magnifying power of the microscope would be

(A) 100

(B) 150

(C) 200

(D) 250

(C) 200

10. Answer the following :

3

- (a) A convex lens produces a real image of an object. If the upper half of the lens is covered with a black paper, then how will the image be affected ?
- (b) The time difference between the apparent sunrise and the actual sunrise is about 2 minutes. Explain.
- (c) Modern telescopes use a concave mirror rather than a lens for the objective. Why ?

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10. Answer the following :

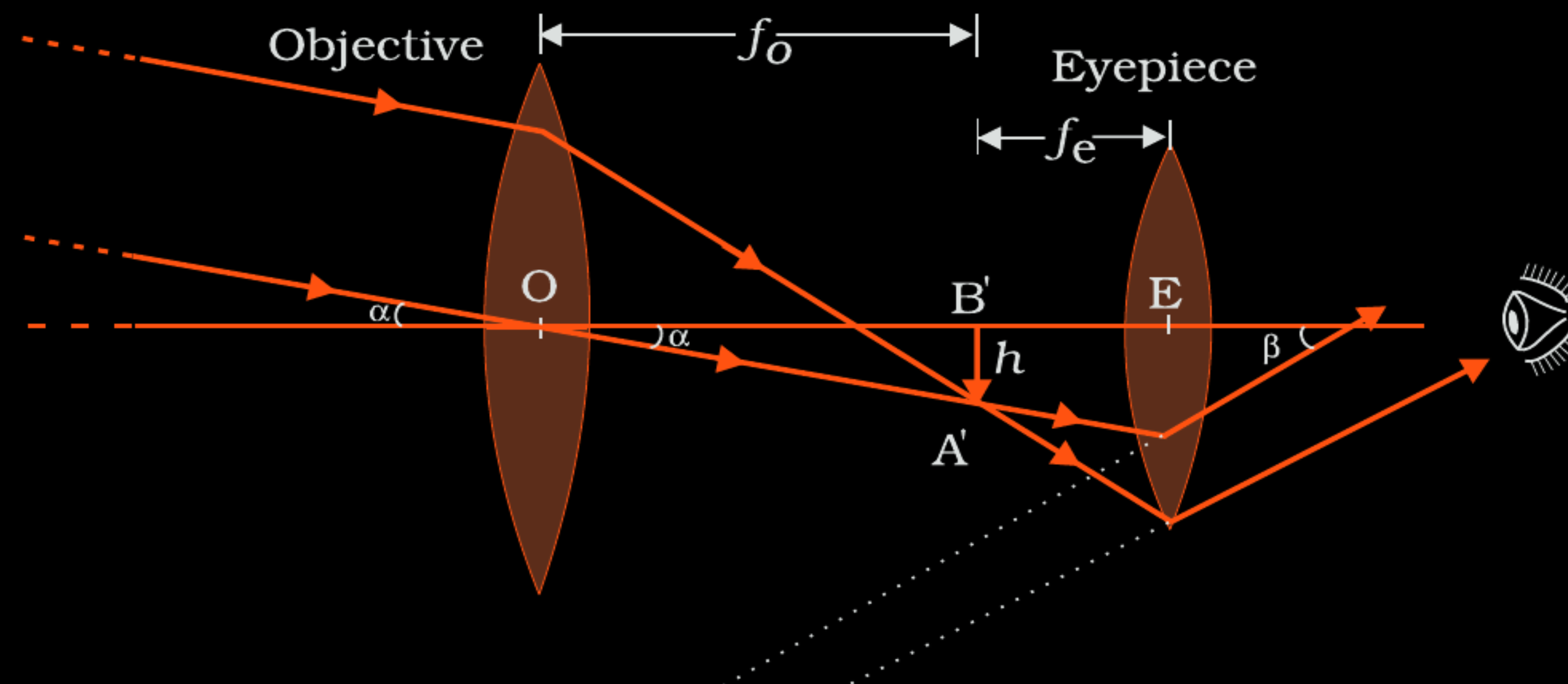
3

- (a) A convex lens produces a real image of an object. If the upper half of the lens is covered with a black paper, then how will the image be affected ?
- (b) The time difference between the apparent sunrise and the actual sunrise is about 2 minutes. Explain.
- (c) Modern telescopes use a concave mirror rather than a lens for the objective. Why ?

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(a) No change in the shape / size and position of the image formed but the intensity of the image decreases	$\frac{1}{2} + \frac{1}{2}$
(b) Due to atmospheric refraction the light coming from the sun keeps bending towards the normal, hence reaches earlier	1
(c) Use of mirror makes the image free from chromatic aberration / improves the quality of image / image is comparatively brighter (any other)	1

7. (a) (i) Draw a labelled ray diagram showing the formation of the image at infinity by an astronomical telescope.
- (ii) A telescope consists of an objective of focal length 150 cm and an eyepiece of focal length 6.0 cm. If the final image is formed at infinity, then calculate :
- (I) the length of the tube in this adjustment, and
- (II) the magnification produced.



1

(ii) Given $f_o = 150 \text{ cm}$, $f_e = 6 \text{ cm}$

(I) Length of the tube $L = f_o + f_e$
 $= 150 + 6$

$$L = 156 \text{ cm}$$

$\frac{1}{2}$

$$(II) \quad m = \frac{f_o}{f_e}$$

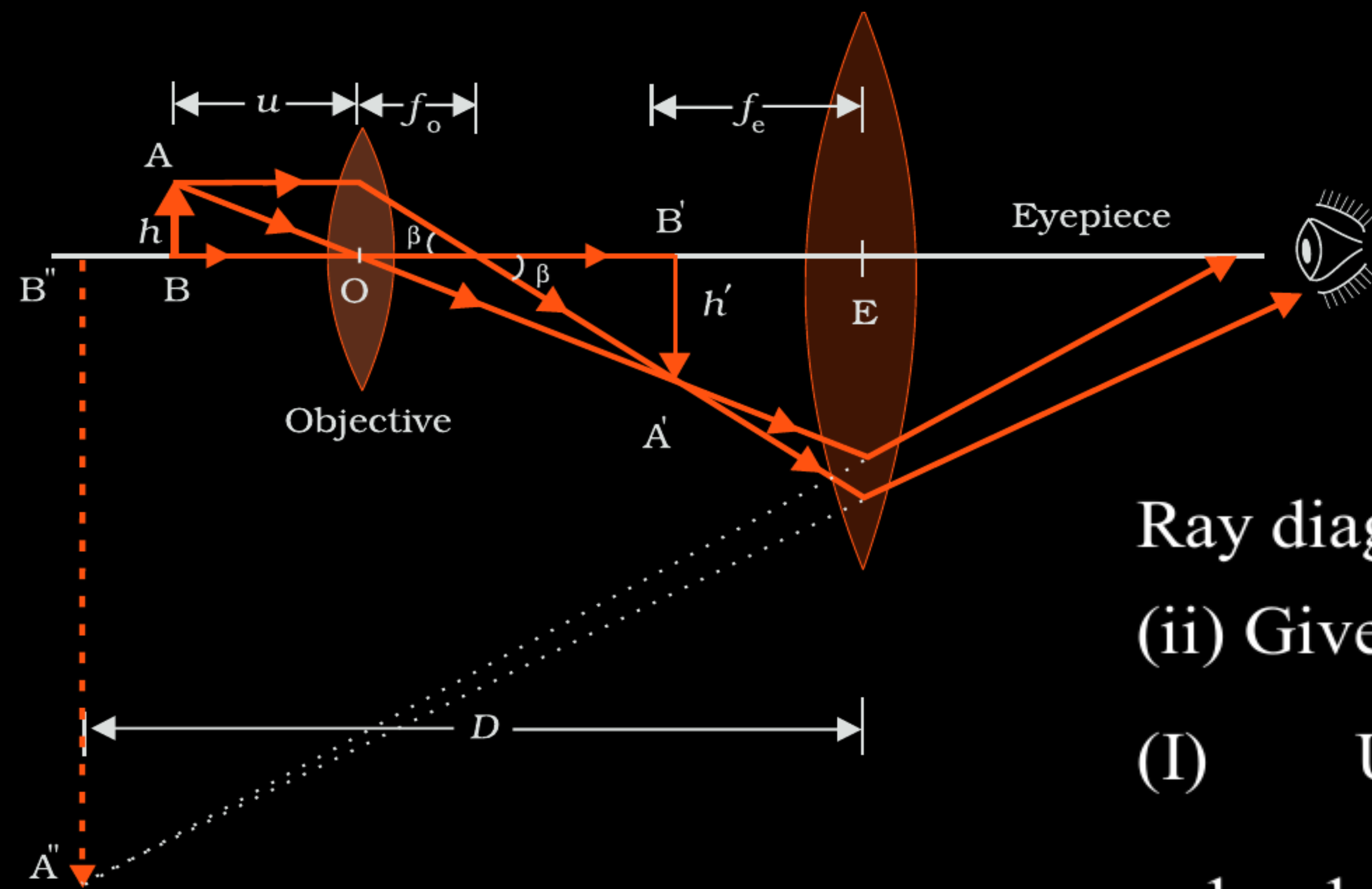
$$\frac{150}{6} = 25$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) (i) Draw a labelled ray diagram showing the formation of the image at least distance of distinct vision by a compound microscope.
- (ii) A small object is placed at a distance of 3.0 cm from a magnifier of focal length 4.0 cm. Find :
- (I) the position of the image formed, and
- (II) the linear magnification produced.



Ray diagram of image formation by a compound microscope

(ii) Given $u = -3 \text{ cm}$ $f = 4 \text{ cm}$

(I) Using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = \frac{1}{-3.0} + \frac{1}{4.0}$$

$$\frac{1}{v} = \frac{-4 + 3}{12} = -\frac{1}{12} \quad v = -12 \text{ cm}$$

(II) Linear magnification $m = \frac{v}{u}$

$$m = \frac{-12}{-3} = 4$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

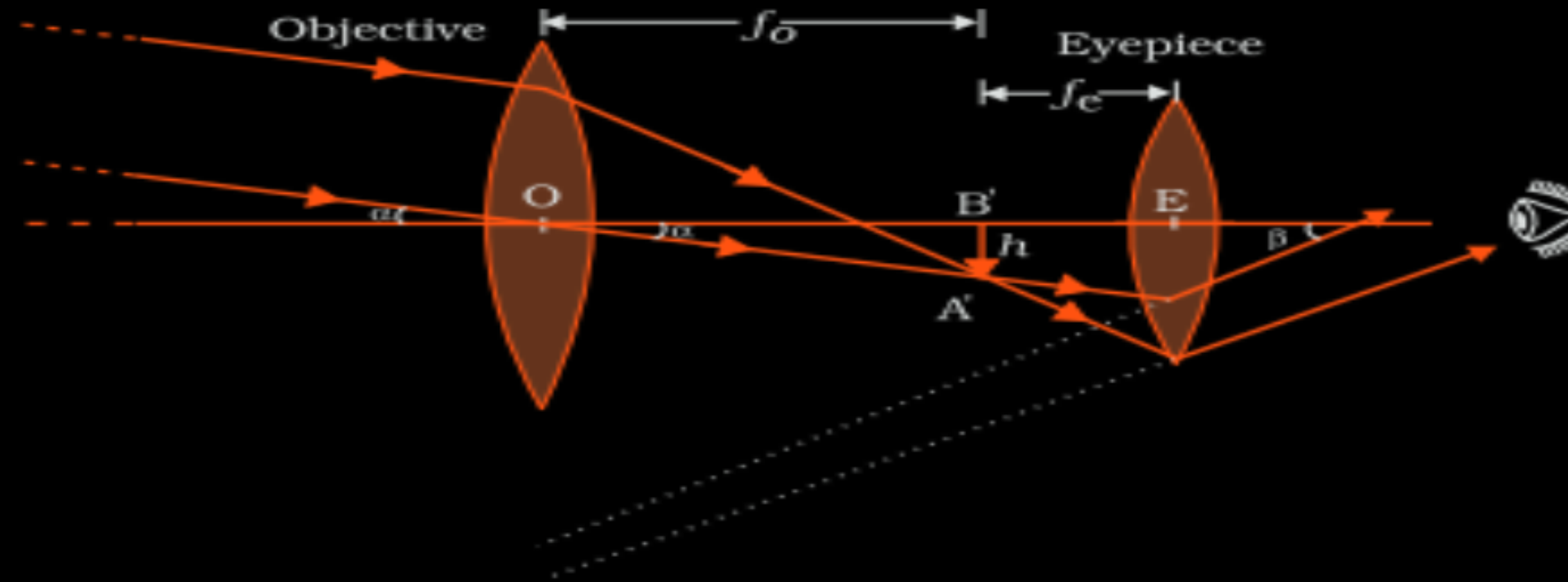
- 9.** (a) Draw a labelled ray diagram showing the formation of an image by an astronomical refracting telescope in normal adjustment. Hence, obtain the expression for its magnifying power.

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9. (a) Draw a labelled ray diagram showing the formation of an image by an astronomical refracting telescope in normal adjustment. Hence, obtain the expression for its magnifying power.

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3



1½

$$m = \frac{\tan \beta}{\tan \alpha} \approx \frac{\beta}{\alpha} \text{ (as } \alpha, \beta \text{ are small angles)}$$

½

$$\approx \frac{h}{f_e} \times \frac{f_o}{h}$$

½

$$= \frac{f_o}{f_e}$$

½

(Note: ½ mark to be deducted if arrows not shown or labelling is not done)

32. An optical instrument uses a lens of power 100 D for objective lens and 50 D for its eyepiece. When the tube length is kept at 25 cm. the final image is formed at infinity.

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- (a) Identify the optical instrument
- (b) Calculate the magnification produced by the instrument.

3

Objective lens with a power of 100 D, has a focal length of 1cm (very short focal length)	
Eye piece, with a power of 50 D, has a focal length of 2cm (short focal length)	1/2
The optical instrument is therefore a compound microscope.	1/2
(Note: Award this one mark if a student writes directly compound microscope without justifying.)	
When the final image is formed at infinity, the magnification of a compound microscope equals	
$m = \left(\frac{L}{f_0} \right) \left(\frac{D}{f_e} \right)$	
$L = 25 \text{ cm}, D = 25 \text{ cm}$	1
$f_0 = 1 \text{ cm}, f_e = 2 \text{ cm}$	
$m = \left(\frac{25}{1} \times \frac{25}{2} \right)$	1/2
$= 312.5$	1/2

19. An astronomical telescope may be a refracting type or a reflecting type. Which of the two produces image of better quality ? Justify your answer. **1**

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19. An astronomical telescope may be a refracting type or a reflecting type. Which of the two produces image of better quality ? Justify your answer.

1

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Reflecting type telescope	$\frac{1}{2}$
Reason/Justification :-	
Mirror have large aperture/high resolving power/ free from chromatic aberration /free from spherical aberration. (Any one)	$\frac{1}{2}$

- 31.** What is the difference in the construction of an astronomical telescope and a compound microscope ? The focal lengths of the objective and eyepiece of a compound microscope are 1.25 cm and 5.0 cm, respectively. Find the position of the object relative to the objective in order to obtain an angular magnification of 30 when the final image is formed at the near point.

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Aperture of telescope objective lens is large whereas aperture of microscope objective is small

1/2

$f_o > f_e$ in telescope
 $f_o < f_e$ in microscope

1/2

[Alternatively, focal length of telescope objective is large whereas focal length of microscope objective is very small]

[Award full 1 mark even if a student writes only one difference]

$$m = m_o \times m_e$$

$$m_e = 1 + \frac{D}{f_e} = 1 + \frac{25}{5} = 6$$

1/2

$$\therefore m_o = \frac{30}{6} = -5$$

1/2

$$m_o = \frac{v}{u} = -5$$

$$v = -5u$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

1/2

$$\frac{1}{1.25} = \frac{1}{5u_o} + \frac{1}{u_o} \quad ; u = -u_o$$

1/2

$$u_o = \frac{6}{5} \times 1.25 = 1.5 \text{ cm}$$

[Alternatively,

$$m_o = \frac{v}{u} = \frac{f_o}{f_o + u_o}$$

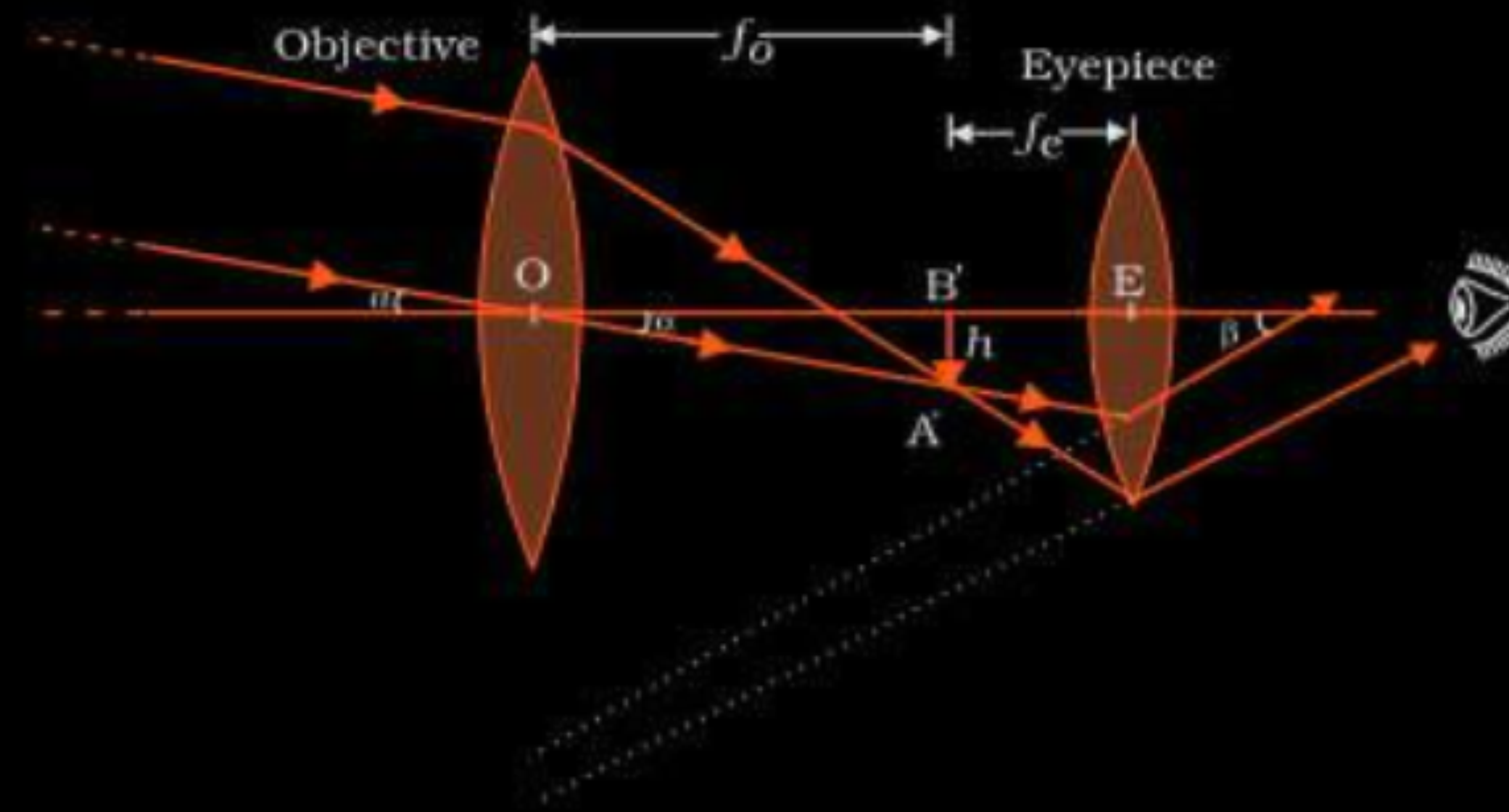
$$-5 = \frac{1.25}{1.25 + u_o}$$

$$-7.5 = 5u_o$$

$$u_o = -1.5 \text{ cm} \quad (\text{for last } \frac{1}{2} \text{ mark})]$$

- 37.** (a) Draw the ray diagram of an astronomical telescope when the final image is formed at infinity. Write the expression for the resolving power of the telescope.
- (b) An astronomical telescope has an objective lens of focal length 20 m and eyepiece of focal length 1 cm.
- (i) Find the angular magnification of the telescope.
- (ii) If this telescope is used to view the Moon, find the diameter of the image formed by the objective lens. Given the diameter of the Moon is 3.5×10^6 m and radius of lunar orbit is 3.8×10^8 m.

a)



Resolving power of telescope = $\frac{D}{1.22\lambda}$

b) (i) Angular magnification $m = \frac{\beta}{\alpha} = \frac{f_o}{f_e} = \frac{20m}{10^{-2}m} = 2000$

(ii)

$$\frac{D}{d} = \frac{x}{f_o}$$

$$d = \frac{D f_o}{x} = \frac{3.5 \times 10^6 \times 20}{3.8 \times 10^8} = .18m$$

2

1

1

$\frac{1}{2}$

$\frac{1}{2}$

- (a) Draw a labelled ray diagram of compound microscope, when final image forms at the least distance of distinct vision.
- (b) Why is its objective of short focal length and of short aperture, compared to its eyepiece ? Explain.
- (c) The focal length of the objective is 4 cm while that of eyepiece is 10 cm. The object is placed at a distance of 6 cm from the objective lens.
 - (i) Calculate the magnifying power of the compound microscope, if its final image is formed at the near point.
 - (ii) Also calculate length of the compound microscope.

The magnifying power of compound microscope

(b)

$$m = m_o \times m_e = \frac{L}{f_o} \times \left(1 + \frac{D}{f_e}\right)$$

To have high magnifying power and high resolution, the focal length of the objective and its aperture should be short.

Focal length of eyepiece is comparatively greater than the objective so that image formed by objective lens may form within the focal length of eyepiece and the final magnified image may be formed.

Aperture in short for higher resolution.

(c)

$$u_o = -6\text{cm}$$

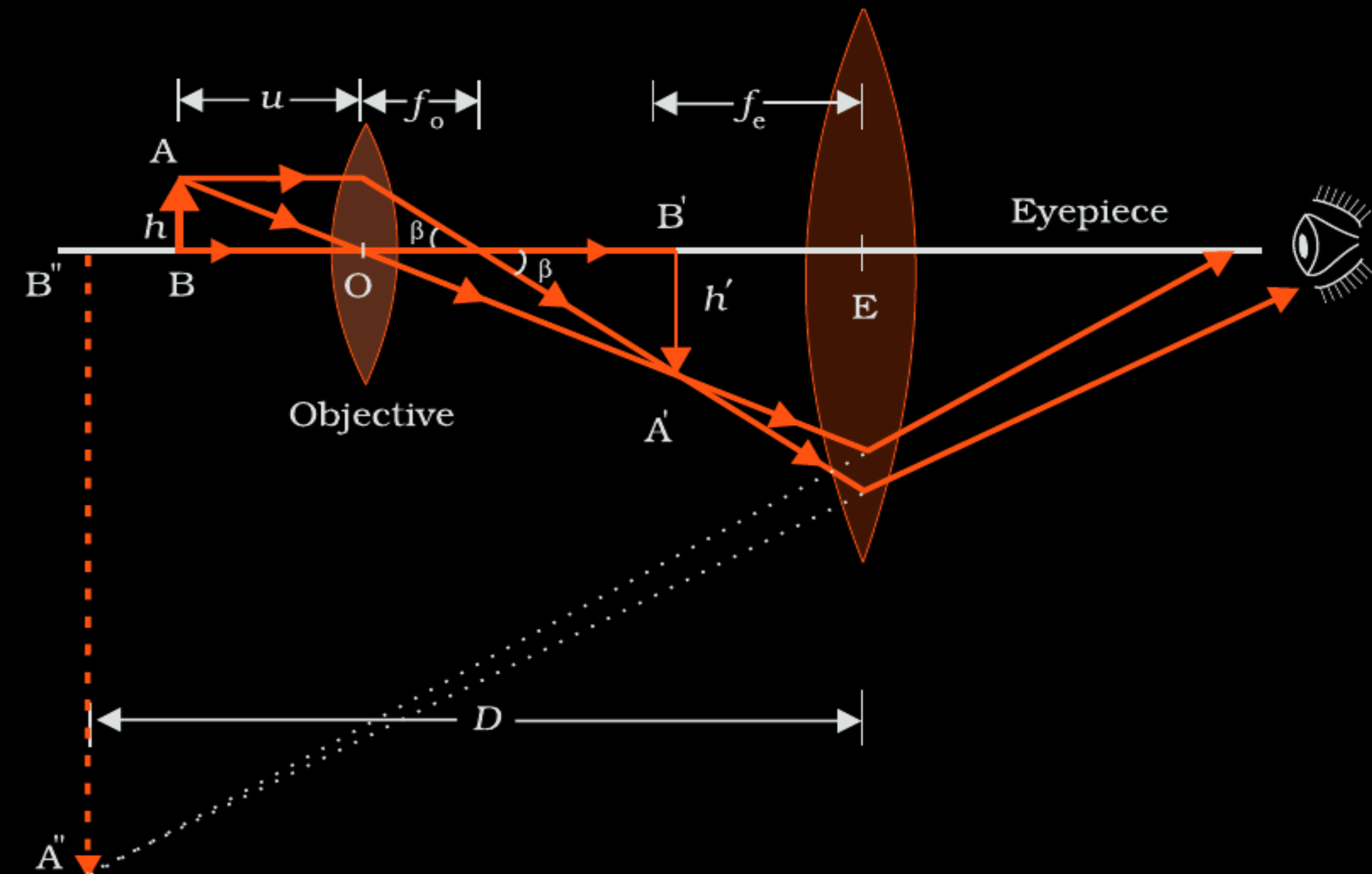
for objective lens

$$\frac{1}{f_o} = \frac{1}{v_o} - \frac{1}{u_o}$$

$$\frac{1}{v_o} = \frac{1}{f_o} + \frac{1}{u_o} = \frac{1}{4} + \frac{1}{-6}$$

$$v_o = 12\text{cm}$$

Since for eyepiece $v_e = -D = -25\text{cm}$



1

1

$$\frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{-25} - \frac{1}{10}$$

$$u_e = -7.14 \text{ cm}$$

(i) For magnifying power of compound microscope

$$m = \frac{v_o}{u_o} \left(1 + \frac{D}{f_e} \right)$$

$$m = \frac{12}{6} \left(1 + \frac{25}{10} \right) = 7$$

(ii) Length of Microscope

$$L = v_o + |u_e| = 12 + 7.14$$

$$L = 19.14 \text{ cm}$$

1

1

- (a) With the help of a labelled ray diagram, explain the construction and working of a Cassegrain reflecting telescope.
- (b) An amateur astronomer wishes to estimate roughly the size of the Sun using his crude telescope consisting of an objective lens of focal length 200 cm and an eyepiece of focal length 10 cm. By adjusting the distance of the eyepiece from the objective, he obtains an image of the Sun on a screen 40 cm behind the eyepiece. The diameter of the Sun's image is measured to be 6.0 cm. Estimate the Sun's size, given that the average Earth-Sun distance is 1.5×10^{11} m.

It consists for large concave (primary) paraboloidal mirror having in its central part a hole. There is a small convex (secondary) mirror near the focus of concave mirror. Eye pieces if placed near the hole of the concave mirror .

The parallel rays from distance object are reflected by the large concave mirror . These rays fall on the convex mirror which reflects these rays outside the hole. The final magnified image in formed.

(b) For eyepiece.

$$\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$$

$$\text{or } \frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e} = \frac{1}{40} - \frac{1}{10}$$

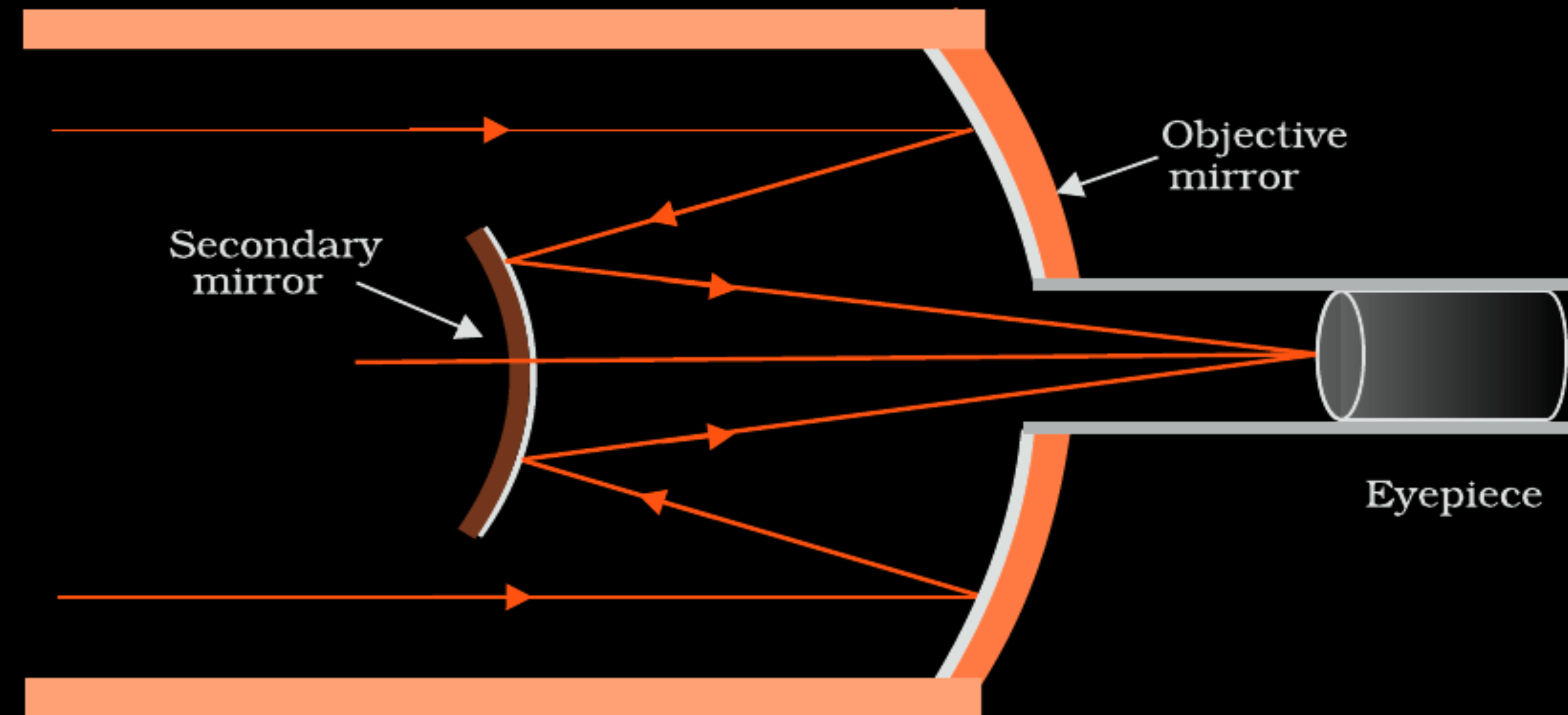
$$u_e = \frac{40}{3} \text{ cm}$$

Magnification produced by eye pieces is

$$m_e = \frac{v_e}{|u_e|} = \frac{40}{40/3} = 3$$

Diameter of the image formed by the objective is

$$d = 6/3 = 2 \text{ cm}$$



1

1/2

1/2

1/2

If D be the diameter of the SUN then the angle subtended by it on the objective will be

$$\alpha = \frac{D}{1.5 \times 10^{11}} \text{ rad}$$

Angle subtended by the image at the objective

= angle subtended by the SUN

$$\therefore \alpha = \frac{\text{size of image}}{f_o} = \frac{2}{200} = \frac{1}{100} \text{ rad.}$$

$$\therefore \frac{D}{1.5 \times 10^{11}} = \frac{1}{100}$$

or

$$D = 1.5 \times 10^9 \text{ m}$$

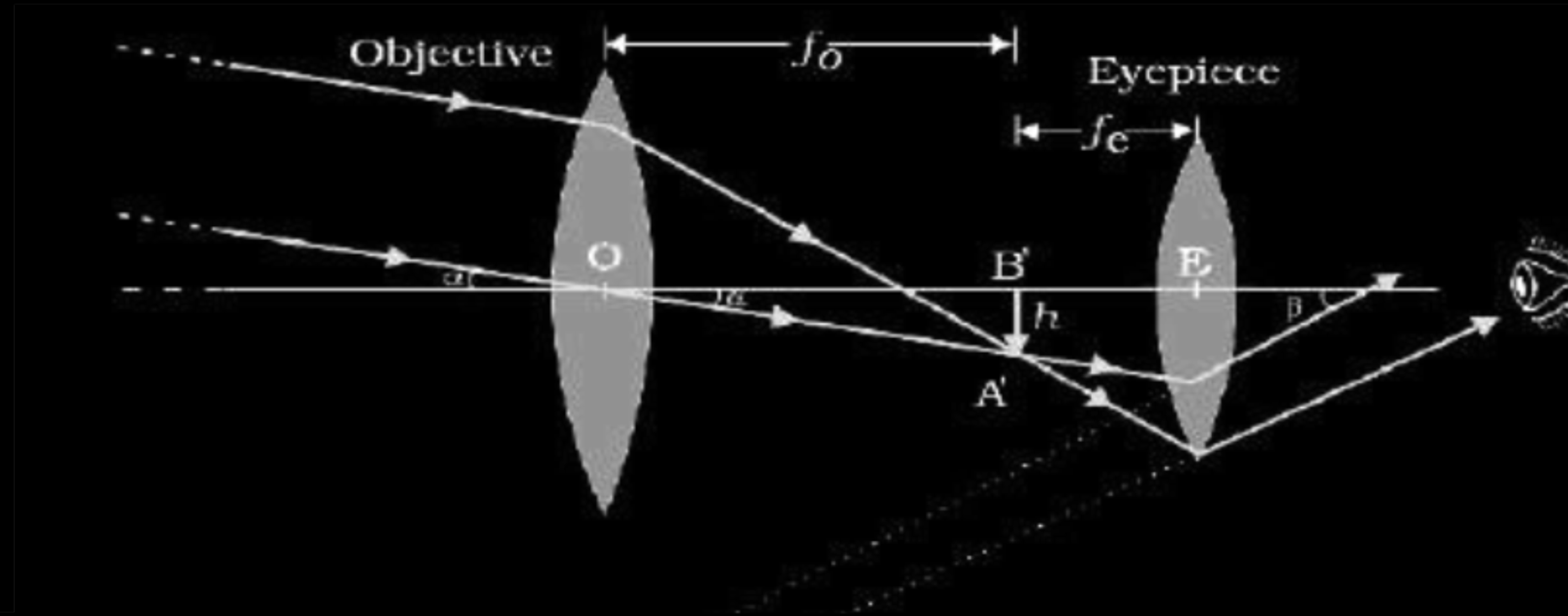
$\frac{1}{2}$

Draw a ray diagram to show the image formation of a distant object by a refracting telescope. Write the expression for its angular magnification in terms of the focal lengths of the lenses used. State the important considerations required to achieve large resolution and their consequent limitations.

CBSE 2019

Draw a ray diagram to show the image formation of a distant object by a refracting telescope. Write the expression for its angular magnification in terms of the focal lengths of the lenses used. State the important considerations required to achieve large resolution and their consequent limitations.

CBSE 2019



Angular magnification $m = \frac{-f_o}{f_e}$ or $\frac{f_o}{f_e}$

Important considerations :

For achieving large resolution, the objective of large aperture is required.

Consequent Limitation : Heavy, hence difficult to make and support by their edge / suffers with chromatic aberrations (any one of above)

1½

½

½

½

State the condition under which a large magnification can be achieved in an astronomical telescope.

OR

How does the angle of minimum deviation of a glass prism vary if the incident violet light is replaced by red light ?

CBSE 2019

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$$f_0 \gg f_e$$

Focal length of objective must be much greater than focal length of eyepiece

OR

Angle of minimum deviation decreases.

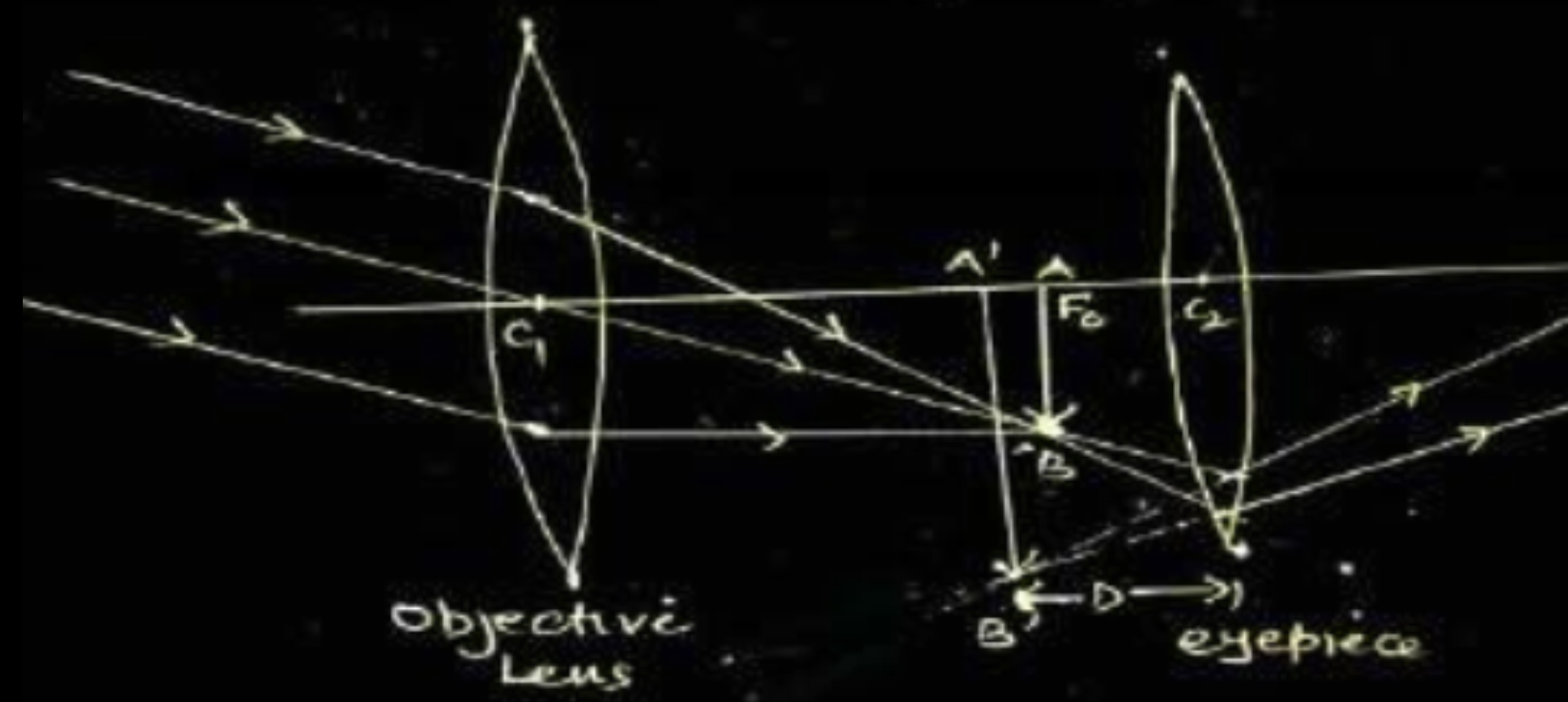
1

1

Draw a labelled ray diagram of an astronomical telescope in the near point adjustment position.

A giant refracting telescope at an observatory has an objective lens of focal length 15 m and an eyepiece of focal length 1.0 cm. If this telescope is used to view the Moon, find the diameter of the image of the Moon formed by the objective lens. The diameter of the Moon is 3.48×10^6 m, and the radius of lunar orbit is 3.8×10^8 m.

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1½

$$\text{Angular magnification of the telescope} = \frac{f_o}{f_e} = \frac{15}{0.01} = 1500$$

½

$$\text{For objective lens, } \tan \alpha = \frac{3.48 \times 10^6}{3.8 \times 10^8}$$

$$\text{For eyepiece } \tan \beta = \frac{h_i}{f_e} = \frac{h_i}{10^{-2}}$$

½

$$\begin{aligned} \therefore \text{Magnifying power} &= \frac{\beta}{\alpha} = \frac{\frac{h_i}{10^{-2}}}{\frac{3.48 \times 10^6}{3.8 \times 10^8}} \\ &= \frac{h_i \times 3.8 \times 10^8}{3.48 \times 10^6 \times 10^{-2}} = 1500 \end{aligned}$$

$$h_i = 13.73 \text{ cm}$$

½

Also accept angular magnification of the telescope

$$= \frac{f_o}{f_e} \left(1 + \frac{f_e}{d} \right) = \frac{15}{0.01} \left(1 + \frac{0.01}{0.25} \right) = 1560$$

So, $h_i = 14.29 \text{ cm}$

Define the magnifying power of a compound microscope when the final image is formed at infinity. Why must both the objective and the eyepiece of a compound microscope has short focal lengths ? Explain.

CBSE 2017

Define the magnifying power of a compound microscope when the final image is formed at infinity. Why must both the objective and the eyepiece of a compound microscope has short focal lengths ? Explain.

CBSE 2017

Magnifying power is defined as the angle subtended at the eye by the image to the angle subtended (at the unaided eye) by the object.

(Alternatively: Also accept this definition in the form of formula)

— —

To increase the magnifying power both the objective and eyepiece must have short focal lengths (as — —

1

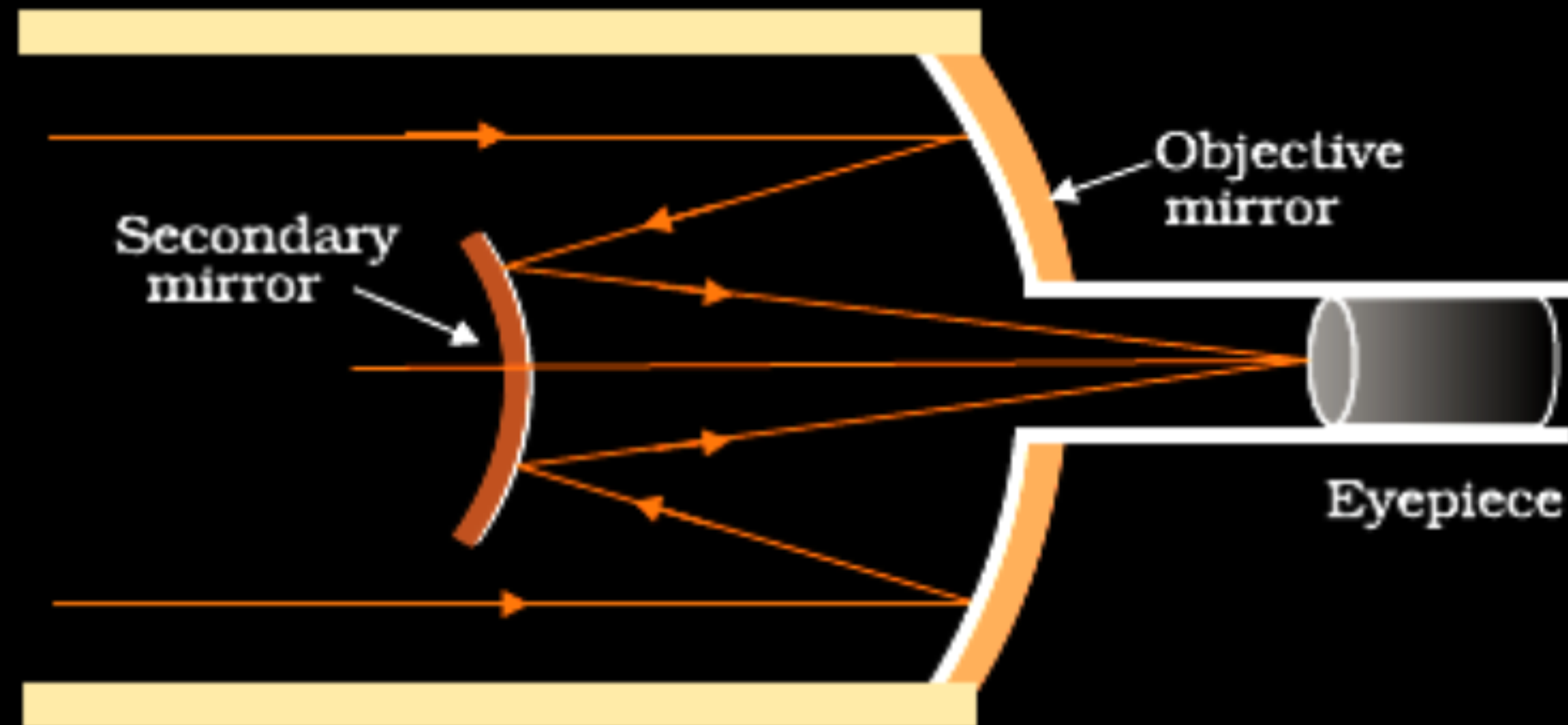
$\frac{1}{2} + \frac{1}{2}$

Draw a schematic ray diagram of reflecting telescope showing how rays coming from a distant object are received at the eye-piece. Write its two important advantages over a refracting telescope.

CBSE 2016

Draw a schematic ray diagram of reflecting telescope showing how rays coming from a distant object are received at the eye-piece. Write its two important advantages over a refracting telescope.

CBSE 2016



- (i) Large gathering power
- (ii) Large magnifying power
- (iii) No chromatic aberration
- (iv) Spherical aberration is also removed
- (v) Easy mechanical support
- (vi) Large resolving power

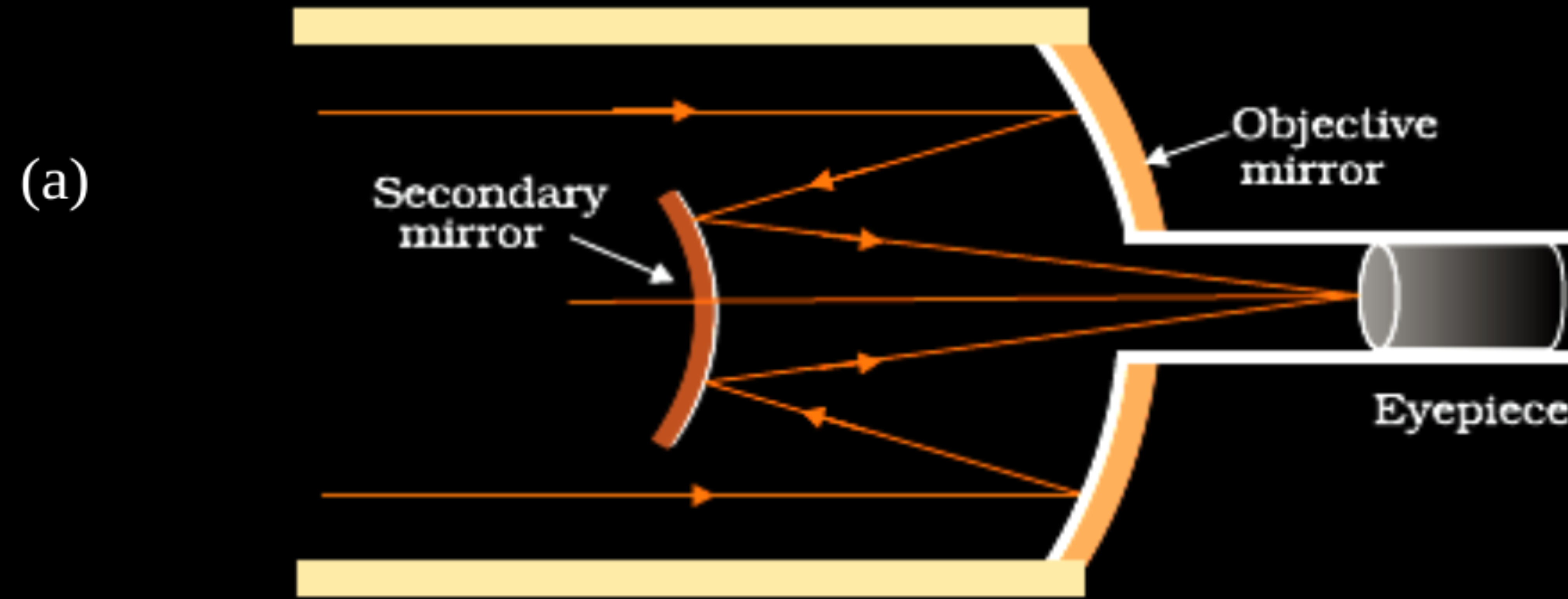
(Any Two)

2

$\frac{1}{2} + \frac{1}{2}$

- (a) Draw a schematic diagram of a reflecting telescope.
- (b) State the advantages of reflecting telescope over refracting telescope.

CBSE 2016



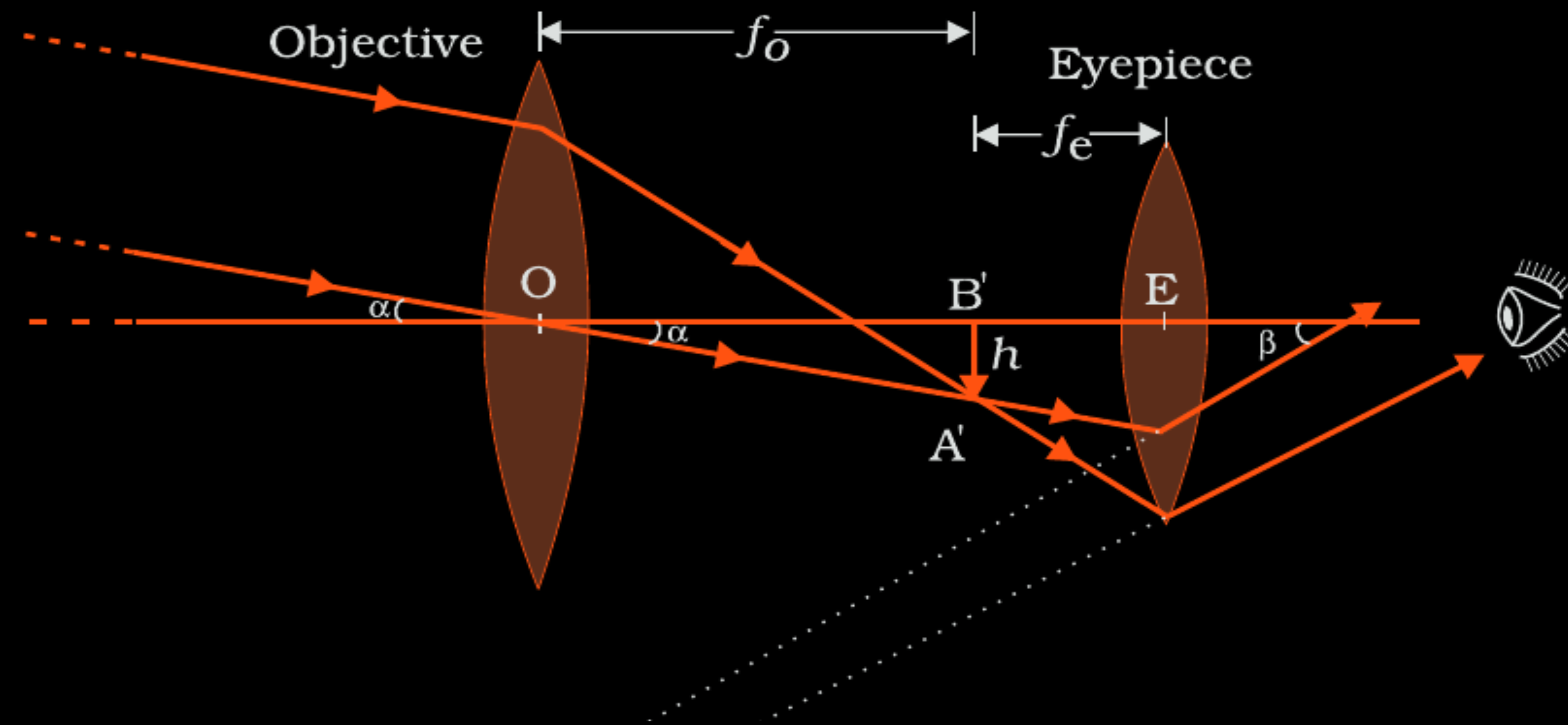
(b) Advantages (any two)

- (i) There is no chromatic aberration in a mirror.
- (ii) Brighter image
- (iii) High resolving Power
- (iv) Large light gathering power
- (v) Large magnifying power

2

$\frac{1}{2} + \frac{1}{2}$

- (a) Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power.
- (b) You are given three lenses of power 0.5 D, 4 D and 10 D to design a telescope.
- (i) Which lenses should be used as objective and eyepiece ? Justify your answer.
- (ii) Why is the aperture of the objective preferred to be large ?



1½

Definition-It is the ratio of the angle subtended at the eye, by the final image, to the angle which the object subtends at the lens, or the eye.

1

b.

i. Objective = .5D

½

½

Eye lens = 10D

This choice would give higher magnification as

$$M = \frac{f_o}{f_e} = \frac{P_e}{P_o}$$

½ + ½

ii. High resolving power/ Brighter image / lower limit of resolution(**any one**)

½